# Decentralized Crypto World Bank

A Blockchain-Based Banking Platform
with AI-Enhanced Security and Assistance

by

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A pre-thesis 1 report submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
**B.Sc. in Computer Science**

Department of Computer Science and Engineering
**BRAC University**
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# Declaration

It is hereby declared that

1. The report submitted is our own original work while completing degree at BRAC University.

2. The report does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.

3. The report does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.

4. We have acknowledged all main sources of help.

**Student's Full Name & Signature:**

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# Approval

The pre-thesis 1 report of final year project titled "Decentralized Crypto World Bank: A Blockchain-Based Banking Platform with AI-Enhanced Security and Assistance" submitted by

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Of Spring, 2026 has been accepted as satisfactory in partial fulfillment of the requirement for the degree of B.Sc. in Computer Science on February 20, 2026.

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# Ethics Statement

This project operates exclusively on public blockchain test networks using test tokens with no real monetary value. No real financial transactions, personal banking data, or user financial records are involved at any stage of development or evaluation. All transaction data used for Artificial Intelligence and Machine Learning (AI/ML) model training and evaluation is either synthetically generated or sourced from publicly available anonymized datasets. The platform prototype does not process Know Your Customer (KYC) or Anti-Money Laundering (AML) data in its current scope, and all wallet addresses used during testing are disposable testnet addresses. We acknowledge the ethical considerations inherent in AI-assisted financial decision-making and have incorporated explainability mechanisms (SHapley Additive exPlanations, or SHAP) to ensure that automated risk assessments remain transparent and auditable by human reviewers. The architectural roadmap incorporates a privacy-preserving Zero-Knowledge Proof (ZKP) identity layer, a separate ZKP-AML circuit against sanctions watchlists, a Decentralized Identifier (DID) anchor backed by W3C Verifiable Credentials, and over-indebtedness controls grounded in 2025 Bangladesh microfinance evidence; the ethical implications of these design choices are evaluated throughout this document and will be re-examined empirically in the final thesis.

# Abstract

Global development finance relies on multilayered institutional structures to distribute capital across borders and communities, yet these systems often struggle with fragmented transparency, procedural friction, and uneven access to credit. Although decentralized finance has demonstrated that programmable infrastructures can automate lending and enhance auditability, existing models largely employ flat architectures that do not reflect institutional hierarchies or structured governance. As a result, there remains limited exploration of how tiered financial systems might be represented within decentralized environments while preserving oversight and adaptability. Here we present *Crypto World Bank*, a proposed architecture that models a multi-tier institutional lending system on programmable blockchain infrastructure. We specify how hierarchical capital flows, role-based governance, and data-informed risk analytics can be coordinated within a unified smart contract environment, enabling transparent state visibility alongside adaptive decision support. The architecture covers six functional domains – deposit mobilization, credit allocation, payment settlement, risk intermediation, liquidity management, and ancillary financial services – within a four-tier hierarchical governance structure, and adds five contributions that target the institutional-DeFi gap identified by recent industry analysis: a stablecoin-first loan denomination model aligned with MiCA and the U.S. GENIUS Act; a liquidation engine with on-chain enforcement; a unified AI risk pipeline wired through a commit–reveal oracle into the lending workflow; a dual zero-knowledge identity layer covering both KYC and AML; and a soulbound credit-passport token that makes repayment history portable across compatible protocols. This work is at the **design and specification** stage: core lending workflow contracts are partially scaffolded on a public testnet, while extended banking modules, a unified fine-tuned AI assistant, federated fraud intelligence, formal verification, and production deployment are planned for subsequent thesis phases.

**Keywords:** Blockchain, Decentralized Finance, Hierarchical Architecture, Financial Inclusion, Smart Contracts, AI-Augmented Governance, Zero-Knowledge Proofs, Group Lending, Deposit Mobilization, Reserve Management, Soulbound Tokens, Federated Learning, Bangladesh Microfinance

# Document Status (Pre-thesis 1)

**Stage:** Architecture and software-engineering specification with a partial testnet prototype. No production deployment, no real-money operations, no production KYC/AML processing.

**Pre-thesis 1 emphasis:** Chapter 1 (Course Outcome CO1) and Chapter 2 (CO9) per the department rubric. Chapters 3–5 establish the architecture, methodology, and feasibility that the final thesis will implement.

**Format target:** BRAC University thesis front matter (this section through the List of Abbreviations) followed by ACM-style body conventions: booktabs tables, BibTeX with `ACM-Reference-Format`, `\Description{}` alt text on every figure, plain black headings, no decorative row shading.

**New in v14 (vs. v13):** Stablecoin-first loan design with MiCA/GENIUS mapping; liquidation engine, SavingsVault and FixedDeposit modules; defense-in-depth security framing; tiered KYC + Account Abstraction onboarding; AI–oracle commit–reveal wiring; ZKP-AML beside ZKP-KYC; W3C DID/VC layer; soulbound credit passport; federated learning specification; Foundry invariant + Certora verification plan; 43 redesigned mermaid figures with a shared visual language; PRISMA-style literature review; consolidated tables; refreshed Bangladesh microfinance data.

# Dedication

*Dedicated to our families for their unwavering support throughout our academic journey.*

# Acknowledgment

We would like to express our sincere gratitude to our panel members for their guidance and support throughout this project. We also thank our supervisor, Mr. Annajiat Alim Rasel, Senior Lecturer at the Department of Computer Science and Engineering, BRAC University, for his guidance and feedback that shaped the diagram and table consolidation, the literature review structure, and the addition of the institutional-DeFi positioning. Finally, we thank the Department of Computer Science and Engineering at BRAC University for providing us with the resources and academic environment to pursue this work.

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# List of Formulas

**F.1 Utilization Rate ($U$):** $U = \frac{L}{L+B}$ (lending utilization; $L$ is lent amount, $B$ is available liquidity)

**F.2 Collateral Ratio (CR):** $CR = \frac{C}{D}$ (collateral-to-debt ratio)

**F.3 Loan-to-Value (LTV):** $LTV = \frac{\text{Loan Amount}}{\text{Collateral Value}}$

**F.4 APR (simple):** $APR = r_{\text{period}} \times N$

**F.5 Compound Interest (Savings):** $A = P\left(1 + \frac{r}{n}\right)^{nt}$

**F.6 EMI (Installment Payment):** $EMI = \frac{P \cdot r \cdot (1+r)^n}{(1+r)^n - 1}$

**F.7 Health Factor (HF):** $HF = \frac{C \times LT}{D}$; $HF < 1.0$ triggers `LiquidationEngine`

**F.8 Reserve Ratio (RR):** $RR = \frac{\text{Reserves}}{\text{Total Deposits}}$

**F.9 Capital Adequacy Ratio (CAR):** $CAR = \frac{\text{Tier 1 Capital}}{\text{Risk-Weighted Assets}}$

**F.10 Money Multiplier ($M$):** $M = \frac{1}{RR}$

**F.11 FX Conversion:** $\text{Amount}_B = \text{Amount}_A \times \left(\frac{P_A}{P_B}\right)$

**F.12 Group Loan Individual Share:** $\text{Share}_i = \frac{\text{LoanTotal}}{N_{\text{members}}}$

**F.13 On-Chain Credit Score:** $CS = \sum_i w_i \cdot \text{feature}_i$

**F.14 Isolation Forest Anomaly Score:** $s(x,n) = 2^{-\frac{E(h(x))}{c(n)}}$

**F.15 Random Forest Fraud Probability:** $P(\text{fraud} \mid x) = \frac{1}{T}\sum_{t=1}^{T} f_t(x)$

**F.16 Platform Net Interest Allocation:** $\text{NetInterest} = \text{InterestCollected} - \text{DepositorYield} - \text{InsuranceAllocation}$

**F.17 SHAP (Shapley value):** $\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!}\left[f(S \cup \{i\}) - f(S)\right]$

**F.18 Composite Risk Score:** $R = w_1 \cdot P(\text{fraud} \mid x) + w_2 \cdot s(x,n)$, with default $w_1 = 0.7, w_2 = 0.3$

**F.19 Rolling Borrowing Limit:** $L_{\text{rem}}(t) = L_{\text{tier}} \cdot k_{\text{loyalty}} - \sum_{i:\, t_i \in [t-180d,t]} a_i$

**F.20 FedAvg Aggregation:** $\theta^g_{t+1} = \sum_{k=1}^{K} \frac{n_k}{n}\theta^k_t$ (server-side averaging of $K$ Local Bank parameter updates)

# List of Abbreviations

**AA:** Account Abstraction (ERC-4337)

**ALM:** Asset-Liability Management

**AI / AI/ML:** Artificial Intelligence / Artificial Intelligence and Machine Learning

**AML:** Anti-Money Laundering

**APR:** Annual Percentage Rate

**API:** Application Programming Interface

**BRAC:** Bangladesh Rural Advancement Committee

**CAR:** Capital Adequacy Ratio

**CBDC:** Central Bank Digital Currency

**CCIP:** Cross-Chain Interoperability Protocol

**CR:** Collateral Ratio

**DAO:** Decentralized Autonomous Organization

**DeFi:** Decentralized Finance

**DID:** Decentralized Identifier (W3C)

**DON:** Decentralized Oracle Network

**DSCR:** Debt Service Coverage Ratio

**EER:** Enhanced Entity Relationship

**ERD:** Entity Relationship Diagram

**EMI:** Equated Monthly Installment

**EVM:** Ethereum Virtual Machine

**FL:** Federated Learning

**FX:** Foreign Exchange

**GNN:** Graph Neural Network

**GPC:** Global Public Goods (UNDP terminology)

**HF:** Health Factor

**IPFS:** InterPlanetary File System

**KYC:** Know Your Customer

**L2:** Layer 2

**LLM:** Large Language Model

**LTV:** Loan-to-Value Ratio

**MFI:** Microfinance Institution

**MiCA:** Markets in Crypto-Assets (EU Regulation 2023/1114)

**ML:** Machine Learning

**MVP:** Minimum Viable Product

**NIM:** Net Interest Margin

**OFAC:** Office of Foreign Assets Control (US sanctions list)

**PoS:** Proof of Stake

**PRISMA:** Preferred Reporting Items for Systematic Reviews

**QRC:** QR code

**QLoRA:** Quantized Low-Rank Adaptation

**RAG:** Retrieval-Augmented Generation

**RR:** Reserve Ratio

**RWA:** Real-World Asset

**SBT:** Soulbound Token

**SDLC:** Software Development Life Cycle

**SHAP:** SHapley Additive exPlanations

**SOFR:** Secured Overnight Financing Rate

**SSE:** Server-Sent Events

**SSI:** Self-Sovereign Identity

**SWC:** Smart Contract Weakness Classification

**TAM/SAM/SOM:** Total / Serviceable / Serviceable Obtainable Market

**TVL:** Total Value Locked

**U:** Utilization

**UUPS:** Universal Upgradeable Proxy Standard (ERC-1822)

**VC:** Verifiable Credential (W3C)

**XAI:** Explainable AI

**ZKP:** Zero-Knowledge Proof

# Chapter 1

# Introduction

The Crypto World Bank (CWB) is not a lending protocol. It is a complete institutional banking architecture specified for implementation on programmable blockchain infrastructure. Where existing decentralized finance protocols address isolated financial functions – Aave handles collateralized lending [Aave(2022)], Uniswap handles exchange, MakerDAO issues stablecoins – CWB integrates the full spectrum of banking functions within a single, hierarchically governed platform. These functions include deposit mobilization, credit allocation, installment-based repayment, interbank liquidity management, foreign exchange facilitation, group lending, reserve enforcement, and risk-based governance – coordinated across four institutional tiers that mirror the structural logic of real-world development finance [Ocampo and Gallagher(2024)].

The system is designed to serve three categories of participants simultaneously. At the global level, it functions as a programmable central reserve institution, managing capital allocation across national jurisdictions with transparent reserve ratios enforced by smart contract code rather than periodic audit [Werner et al.(2022)]. At the national and regional level, it functions as a commercial and development bank, channelling capital to local lending institutions and providing institutional borrowers with structured credit facilities. At the individual level, it functions as a retail bank, offering savings products, group loans, installment personal lending, and digital payment services to end customers, including the 1.4 billion adults excluded from formal financial systems [World Bank(2021)]. The full source code, smart contracts, and supporting material are available in the project repository [Juboraz and Ahmed(2026)].

## 1.1 Background

### 1.1.1 Financial inclusion in 2025

The latest Bangladesh Bank *Special Publication SP2025-02* (June 2025) reports that female-to-male account parity was achieved by March 2025 (49% / 49%), and female-owned deposit accounts grew from 33.4 million in 2019 to 55.3 million in 2024 [Bangladesh Bank(2025b)]. BRAC's 2025 microfinance data shows 89% of microloans go to women and 92% of those borrowers report household income increases [BRAC(2025)]. These are more current and country-specific than the 2021 Findex headline figure of 1.4 billion unbanked adults [World Bank(2021)] that the project's earlier drafts relied on; they are also the empirical anchor for designing the platform's solidarity-group lending tier around women-led borrower groups.

Despite this progress, two structural gaps remain. First, Bangladesh has 724 licensed microfinance institutions serving 41.56 million accounts [NUS Economics Society(2025)], and over-indebtedness from simultaneous borrowing across institutions is now an active policy concern. Second, the International Finance Corporation places the unmet MSME finance gap in developing countries at USD 5.2 trillion annually [International Finance Corporation(2019)]; the World Bank's Migration and Development Brief 40 records an average remittance cost of 6.2% [World Bank(2024)]. Figure 1.1 consolidates these statistics into a single problem-and-funnel view.

### 1.1.2 Programmable infrastructure as a leapfrog opportunity

The World Economic Forum's 2022 analysis [World Economic Forum(2022)] and the Bank for International Settlements 2024 working paper on DeFi lending [Bank for International Settlements(2024a)] both recognise that programmable financial infrastructures collapse the multi-day correspondent banking chain into seconds and reduce settlement cost by an order of magnitude, while making every transaction auditable. JPMorgan's Kinexys (formerly Onyx) settled over USD 1.5 trillion in tokenised payments by 2025 [JPMorgan Chase(2025)], the World Bank piloted blockchain-based fund tracking for its project portfolio [World Bank(2025)], and the BIS mBridge consortium settled real-value cross-border CBDC transactions at minimum-viable-product stage [Bank for International Settlements(2024c)]. Figure 1.2 contrasts traditional correspondent banking with on-chain settlement and shows the resulting transaction lifecycle in CWB.

## 1.2 Rationale of the Study

Existing decentralised lending protocols deliver remarkable transparency and composability but operate on flat, single-pool architectures that erase the institutional layering present in real-world development finance [Werner et al.(2022), Gudgeon et al.(2020)]. The literature explicitly identifies the absence of governance hierarchies, role separation, and reserve discipline as the barrier that prevents large institutional allocators – multilateral development banks, central banks, regulated commercial banks – from participating in DeFi at scale [Sygnum Bank(2026)]. Bridging the gap between a flat liquidity pool and a tier-aware reserve system is the central technical research opportunity that this thesis addresses.

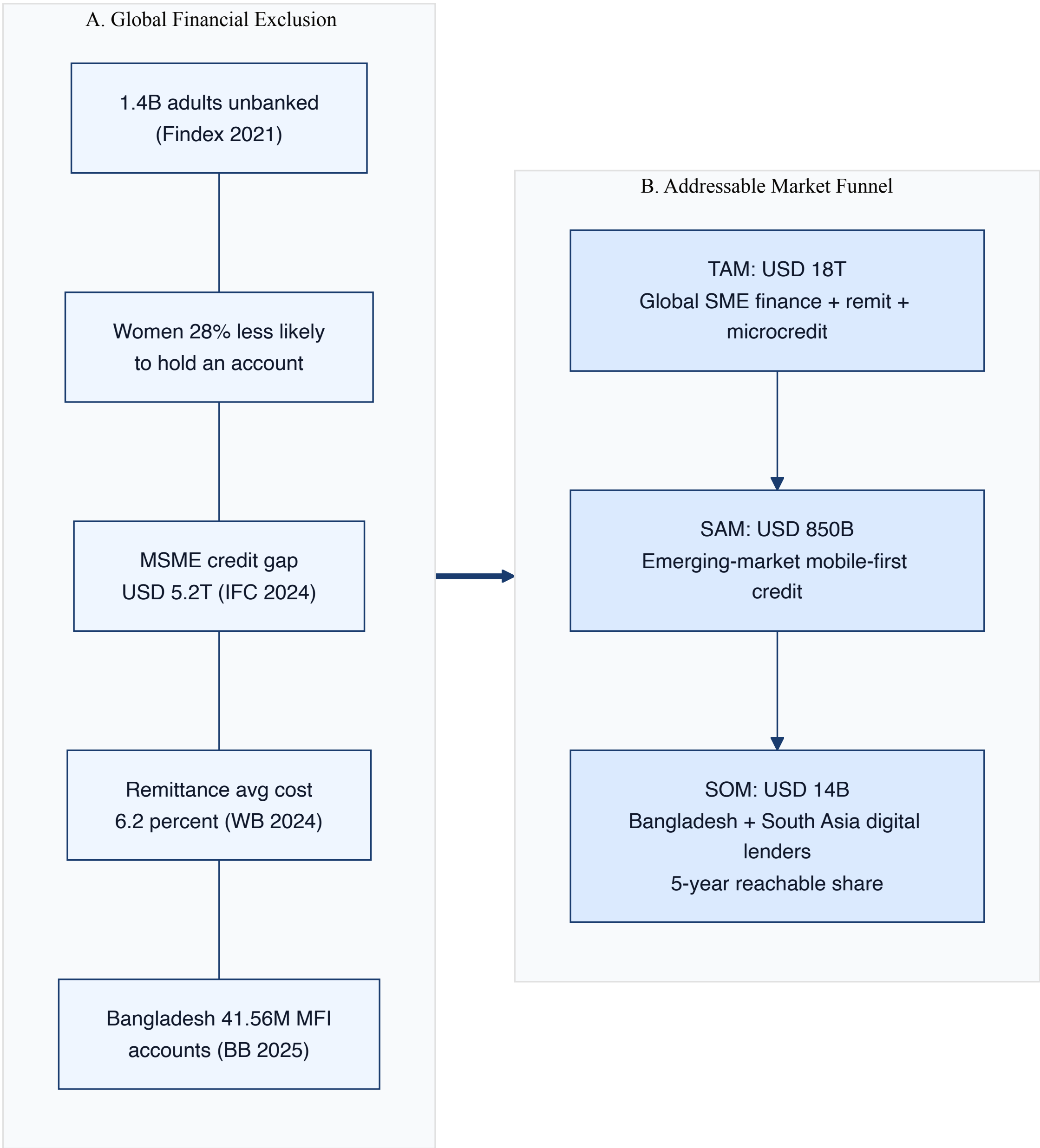



**Figure 1.1:** Global financial exclusion and the addressable-market funnel. Panel A summarizes 2024–2025 inclusion indicators; Panel B applies TAM/SAM/SOM filters to the addressable digital-lending opportunity in Bangladesh and South Asia.

## 1.3 Problem Statement

Sygnum Bank's February 2026 institutional DeFi report [Sygnum Bank(2026)] formulates the gap precisely: **protocols work technically, but no large institutional allocator will participate until legal and regulatory risks are resolved, and no current protocol expresses the hierarchical governance that institutional finance requires**. Compounded with three secondary observations – (a) existing AI-augmented lending stacks present standalone notebooks rather than inte-

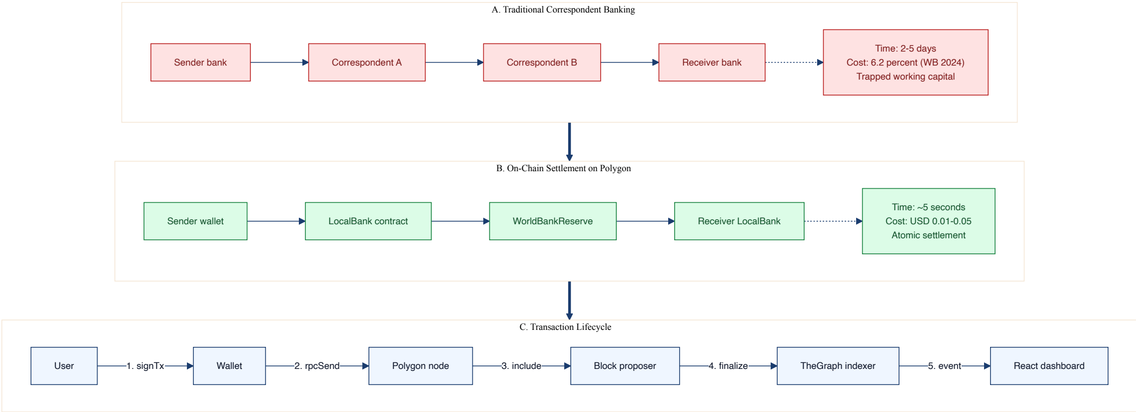



**Figure 1.2:** Correspondent banking versus on-chain settlement on Polygon, with the CWB transaction lifecycle. Panel A is the traditional 2–5 day, 6.2 percent fee, 4-hop chain; Panel B is the 5-second, USD 0.01–0.05 fee, atomic on-chain settlement; Panel C is the CWB transaction lifecycle from wallet signature through block proposer, indexer, and dashboard.

grated risk pipelines [Palaiokrassas et al.(2023), xai(2024)]; (b) ZKP integration in DeFi addresses identity but not sanctions and AML compliance [Piper et al.(2025), zkAML Authors(2025)]; (c) repayment history is locked inside each protocol and not portable across lenders [Buterin et al.(2022)] – the Problem Statement of this work is to specify a blockchain banking architecture that (1) expresses an institutional four-tier hierarchy with on-chain reserve discipline, (2) wires AI risk and explainability into the lending workflow rather than holding them off to the side, (3) extends zero-knowledge compliance from KYC to AML, and (4) makes credit history portable through a soul-bound credit passport.

## 1.4 Objectives

The objectives, ordered for the pre-thesis 1 architecture and software-engineering phase, are:

1. Specify a four-tier hierarchical lending architecture (`WorldBankReserve` → `NationalBank` → `LocalBank` → borrowers) with downward capital allocation and upward surplus repatriation enforced by smart contract code.

2. Define a defense-in-depth security framing covering smart contracts, application, AI/ML, runtime monitoring, and operational layers (Figure 3.1).

3. Specify a stablecoin-first loan denomination model aligned with the EU *Markets in Crypto-Assets* (MiCA) regulation [European Union(2024), Scorechain(2025)] and the U.S. GENIUS Act [United States Congress(2025)].

4. Design a `LiquidationEngine` contract that automates enforcement of the Health Factor formula (F.7) with a 5–8% liquidation bonus, removing reliance on human approvers for liquidation.

5. Wire AI risk scoring (Random Forest + Isolation Forest + SHAP) into the lending workflow through a commit–reveal oracle, with planned migration to Chainlink Functions.

6. Extend the zero-knowledge identity layer with a separate `AMLVerifier` circuit and anchor identity using W3C Decentralized Identifiers and Verifiable Credentials.

7. Specify a soulbound credit-passport token (SBT) that records repayment history and is queryable by any compatible lending protocol.

8. Provide a Bangladesh regulatory pathway through the Regulatory FinTech Facilitation Office sandbox, anchored on FE Circular No. 06 (January 2025) [Bangladesh Bank(2025a)].

## 1.5 Research Questions

- **RQ1.** How can the four-tier institutional lending hierarchy be expressed in smart contract code such that downward allocation, upward surplus repatriation, and reserve invariants are enforced without trusted intermediation?

- **RQ2.** How does wiring an off-chain AI risk pipeline through a commit–reveal oracle affect end-to-end loan latency, gas cost, and the auditability of automated decisions?

- **RQ3.** What additional compliance coverage is gained by extending zero-knowledge identity from KYC to AML against a dynamically updated sanctions Merkle root, and at what gas cost?

- **RQ4.** What is the minimum regulatory pathway through which a stablecoin-denominated, hierarchically governed CWB pilot can operate within the existing Bangladesh and EU regulatory frames?

## 1.6 Research Contribution

The principal contributions of pre-thesis 1 are:

1. A formal four-tier hierarchical lending architecture with explicit cross-tier transfer functions and surplus repatriation (Section 3.9, Figure 3.12).

2. A stablecoin-first design with a MiCA + GENIUS Act compliance mapping table (Appendix E).

3. A commit–reveal AI risk oracle wired into the lending workflow with SHAP-based borrower-facing explanations (Section 3.3.3).

4. A dual ZKP compliance gateway covering both identity (KYC) and sanctions (AML) (Section 3.6, Figure 3.6).

5. A soulbound credit-passport token specification (Section 3.6.4, Figure 3.8).

6. A federated learning specification across Local Banks, with FedAvg aggregation at the National Bank tier (Section 4.3).

7. A formal verification plan with two Certora CVL invariants on the reserve contract (Section 4.5).

8. A PRISMA-style literature review across 38+ studies plus 10 newly selected anchor papers (Chapter 2).

9. Consolidated and visually unified architecture diagrams (43 figures with a shared mermaid theme) replacing 46 inconsistent legacy diagrams.

### 1.6.1 Position vs. institutional CBDC settlement rails

The BIS *mBridge* project (China, Hong Kong, Thailand, UAE, Saudi Arabia) reached minimum-viable-product status in mid-2024 and settled real-value cross-border CBDC transactions; *Project Agora* (BIS + seven central banks) explores tokenised commercial-bank deposits on a unified ledger [Bank for International Settlements(2024c), Bank for International Settlements(2024b)]. Neither project provides a lending hierarchy: they are settlement rails. The IMF's 2025 CBDC survey records that 94% of central banks are engaged in CBDC research [International Monetary Fund(2025)]. We therefore position CWB explicitly as a *complementary lending layer that can run on top of such settlement infrastructure*, rather than as an alternative to traditional banking. This re-framing converts the contribution from "an alternative DeFi protocol" into "a composable institutional lending protocol compatible with emerging CBDC infrastructure" – a substantially more defensible and publishable claim.

## 1.7 Blockchain Justification

### 1.7.1 Blockchain fundamentals and the EVM execution model

The platform uses Ethereum-compatible smart contracts deployed on Polygon Amoy. The Ethereum Virtual Machine [Wood(2014)] provides deterministic, replayable execution of programs whose state is replicated across consensus participants. Polygon's Proof-of-Stake consensus delivers ∼5-second block finality and gas costs in the USD 0.01–0.05 range for typical lending operations, making it the practical environment for prototype work that involves frequent contract interactions. Figure 1.3 shows the full stack used by CWB, from Layer 1 network through smart contracts, off-chain storage, AI/ML services, indexing and monitoring, application orchestration, and client surfaces.

## 1.8 Proposed Solution

### 1.8.1 Banking functions of the platform

Table 1.1 maps the six functional domains expected of a deposit-taking development bank to the contracts and services that implement (or are scheduled to implement) each function within CWB.

### 1.8.2 Service delivery across scale

Three categories of participants are served simultaneously: a programmable central reserve institution at Tier 1 (the World Bank role), commercial and development banks at Tier 2 and Tier 3 (National Banks and Local Banks), and retail / group borrowers at Tier 4 (individuals and solidarity groups). Figure 1.4 fixes both the access boundaries and the cross-tier topology in a single diagram, and Figure 1.5 shows the on-chain role tree that enforces those boundaries.

### 1.8.3 Cross-tier lending system

Same-tier lending (e.g., Local-to-Local liquidity), upward lending (a Local Bank asking its parent National Bank for short-term capital), and downward allocation (the World Bank Reserve granting capacity to a National Bank) are all expressed through three contract functions: `allocateToNationalBank`, `allocateToLocalBank`, and `disburseLoanToUser`. The reverse direction, `returnSurplusToParent`, is what makes the model self-regulating: a Local Bank that accumulates collected interest above its reserve floor remits the surplus to its parent National Bank, which in turn returns surplus to the World Bank Reserve. The complete data and control flow is the subject of Section 3.9 and Figure 3.12.

#### Bangladesh regulatory reality

Bangladesh Bank's stance as of 2025 is that crypto-assets remain effectively illegal for retail use, while the central bank's own CBDC feasibility study (initiated 2022, pilot 2024) is ongoing [Ainvest(2025)]. However, the Regulatory FinTech Facilitation Office at the central bank has established a sandbox path for fintech and FE Circular No. 06 (January 2025) on electronic Letter-of-Credit communication [Bangladesh Bank(2025a)] is now in force, establishing precedent for blockchain-based banking communications. The Payment and Settlement System Act 2024 is the second relevant regulatory anchor. CWB's pilot positioning is therefore **sandbox-first, not mainnet-first**: the

**Table 1.1:** Banking capability matrix. Status legend: ✓ implemented on Polygon Amoy testnet; ▷ partial scaffold; ○ designed for the final thesis; × out of scope.

| **Domain** | **Primary contracts / services** | **User-facing surface** | **Pre-thesis 1** | **Final thesis** |
|---|---|---|---|---|
| Credit allocation | LocalBankContract, LoanController | React + AA wallet | ▷ | ✓ |
| Risk intermediation | FastAPI ML + Oracle Relay + SHAP | SHAP RiskExplanation card | ○ | ✓ |
| Reserve management | WorldBankReserve, NationalBankContract | Admin dashboard | ✓ | ✓ |
| Deposit mobilization | SavingsVault, FixedDeposit | Vault dashboard | ○ | ✓ |
| Payment settlement | Native transfers, stablecoin facility | Send-receive widget | ▷ | ✓ |
| Liquidity management | InterBankLendingPool, surplus repatriation | Bank operator view | ○ | ✓ |
| Group lending | SolidarityGroup, GroupLendingPool | Group dashboard | ○ | ✓ |
| Foreign exchange | FXModule + Chainlink Price Feeds | FX widget | ○ | ✓ |
| Identity & compliance | KYCVerifier (ZKP), AMLVerifier (zkAML) | Onboarding wizard | ○ | ✓ |
| Credit history portability | CreditPassportSBT (ERC-5114-like) | Borrower profile | ○ | ✓ |
| Trade finance | TradeFinanceLC (later phase) | Corporate dashboard | × | ○ |

architecture is shaped so a pilot can enter the sandbox via a Local Bank operator with on-chain reserve gating and ZKP-KYC compliance.

## 1.9 Methodology in Brief

The project follows an Agile/Scrum process across three sprints, with the SDLC mapping detailed in Section 4.13. Pre-thesis 1 corresponds to Sprints 1–2 (architecture, specification, partial scaffolding); pre-thesis 2 covers Sprint 3 and the integration of the AI/ML pipeline, the liquidation engine, the SavingsVault, and the zkAML circuit.

The validation strategy uses Foundry invariant + fuzz testing (Section 4.6), Hardhat integration tests against a Polygon Amoy fork, and an optional formal verification pass with the now-open-source Certora Prover [Certora(2025)] on the reserve contract.

## 1.10 Scopes and Challenges

### 1.10.1 Economic sustainability: interest revenue versus gas costs

The platform's net interest allocation (F.16) is positive at the unit-economics level only if gas costs remain below the spread captured by the platform. Polygon's USD 0.01–0.05 average gas cost per loan operation [Tolmach et al.(2024)] sits well below the spread modelled in Chapter 5, so the unit economics are favourable at the prototype scale. Mainnet Ethereum deployment is explicitly deferred.

### 1.10.2 Resistance to institutional capture

Hierarchical systems risk capture by upper-tier participants. CWB's mitigations are (a) role separation enforced by AccessControl with role expiry timestamps; (b) a 24–48 hour TimeLock delay on every governance action; (c) a 3-of-5 Safe multisig at the World Bank Admin role; (d) public Tenderly alerts on every allocation event; and (e) Soulbound credit passports that resist transfer and capture. Section 3.14 expands the governance framework.

### 1.10.3 Cryptocurrency supply constraints and scalability

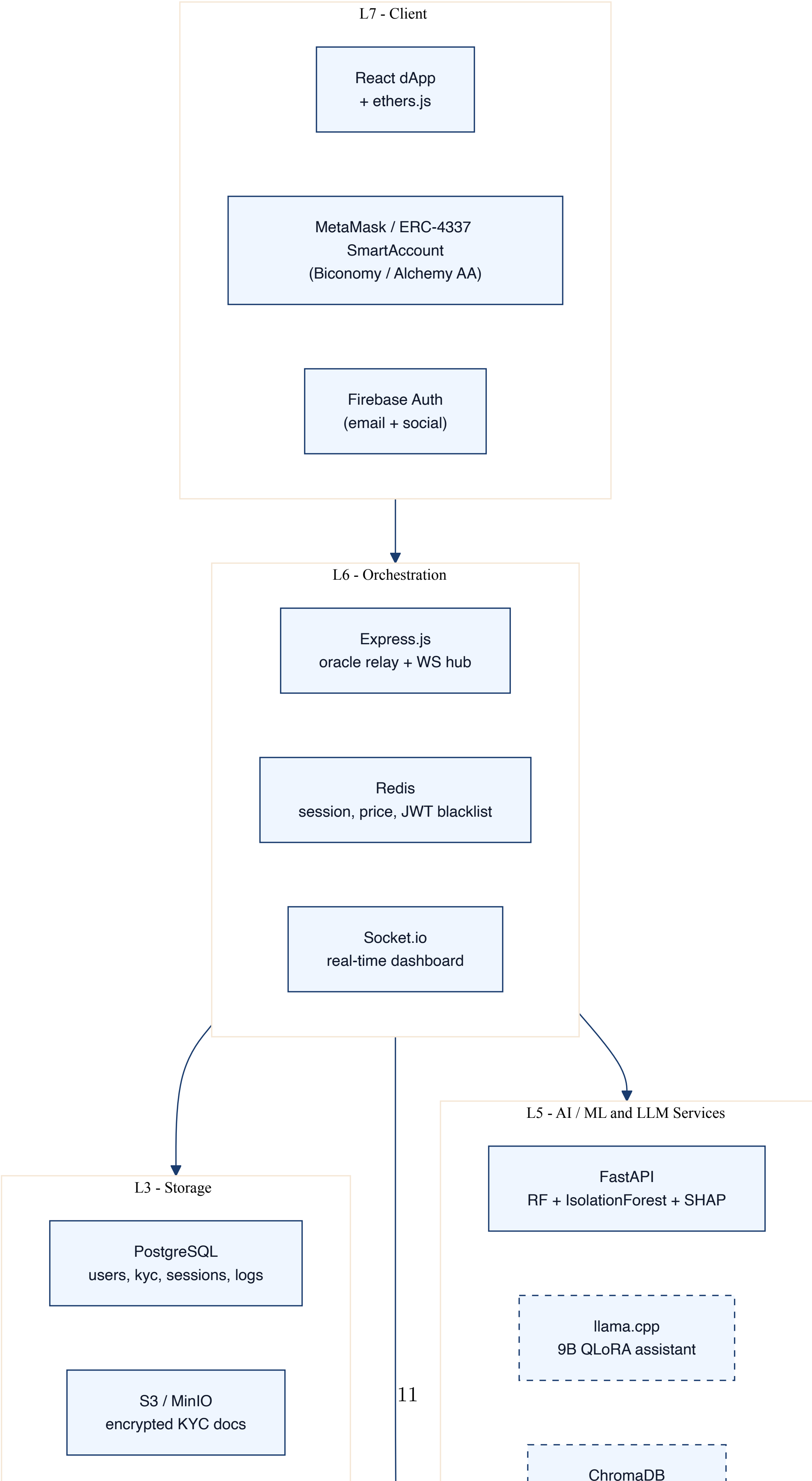
The platform denominates loans in stablecoins – not in ETH – specifically to avoid the catastrophic real-debt doubling a borrower in a 50%-down market would otherwise face [sta(2025)]. MiCA's 1:1 reserve-backing rule for e-money tokens [European Union(2024), Scorechain(2025)] is the regulatory anchor for this design choice.

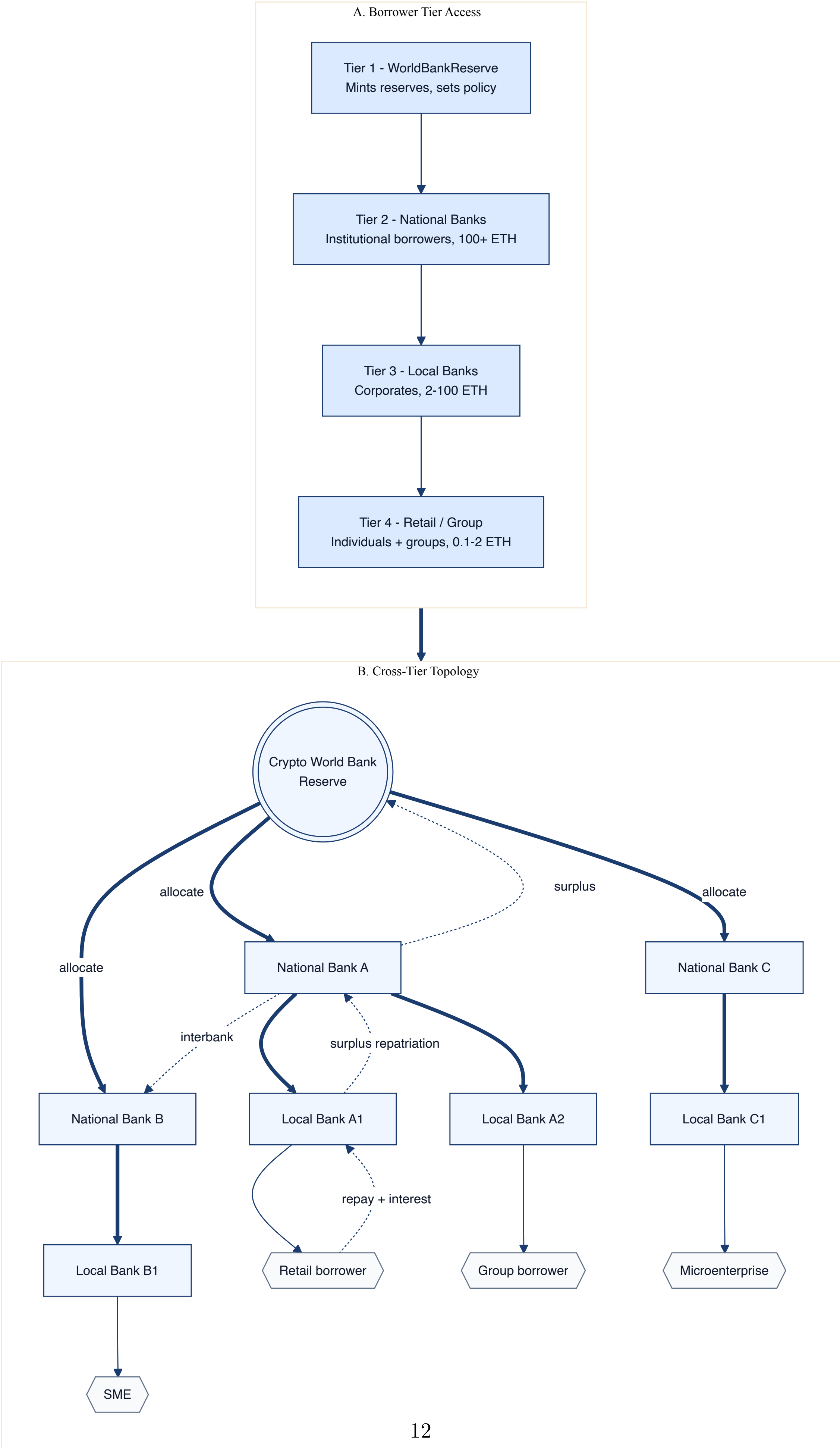
## 1.11 Economic Architecture: Flat vs Hierarchical

Figure 1.6 contrasts the flat single-pool DeFi pattern against the CWB hierarchical lending pattern, situates the architecture within the historical Cantillon-effect distributional analysis, and tracks the interest-rate cascade through the four-tier hierarchy. The cascade resolves to a borrower APR of approximately 8%, two-thirds the rate of comparable Bangladeshi MFIs and one-third that of typical card-credit alternatives.

## 1.12 Market Analysis and Partnership Ecosystem

A summary appears here; the full quantitative analysis is in Chapter 5. The serviceable obtainable market is estimated at USD 14 billion over five years across Bangladesh and South Asia digital-lending segments. The competitor positioning – against Aave, Compound, Maple, Morpho, Goldfinch, MakerDAO, Grameen, BRAC Microfinance, JPM Onyx, and BIS mBridge – places CWB in the previously unoccupied "Multi-tier institutional" quadrant (Figure 5.2, Chapter 5). The phased go-to-market through Sandbox, Pilot, Production, and Regional scale (Figure 5.1, Chapter 5) is anchored on the Bangladesh sandbox path described in Section 1.8.3.


L7 - Client
React dApp
+ ethers.js
MetaMask / ERC-4337
SmartAccount
(Biconomy / Alchemy AA)
Firebase Auth
(email + social)
L6 - Orchestration
Express.js
oracle relay + WS hub
Redis
session, price, JWT blacklist
Socket.io
real-time dashboard
L5 - AI / ML and LLM Services
FastAPI
RF + IsolationForest + SHAP
llama.cpp
9B QLoRA assistant
ChromaDB
L3 - Storage
PostgreSQL
users, kyc, sessions, logs
S3 / MinIO
encrypted KYC docs


A. Borrower Tier Access
Tier 1 - WorldBankReserve
Mints reserves, sets policy
Tier 2 - National Banks
Institutional borrowers, 100+ ETH
Tier 3 - Local Banks
Corporates, 2-100 ETH
Tier 4 - Retail / Group
Individuals + groups, 0.1-2 ETH
B. Cross-Tier Topology
Crypto World Bank
Reserve
allocate
allocate
allocate
surplus
National Bank A
National Bank B
National Bank C
interbank
surplus repatriation
Local Bank A1
Local Bank A2
Local Bank C1
Local Bank B1
repay + interest
Retail borrower
Group borrower
Microenterprise
SME

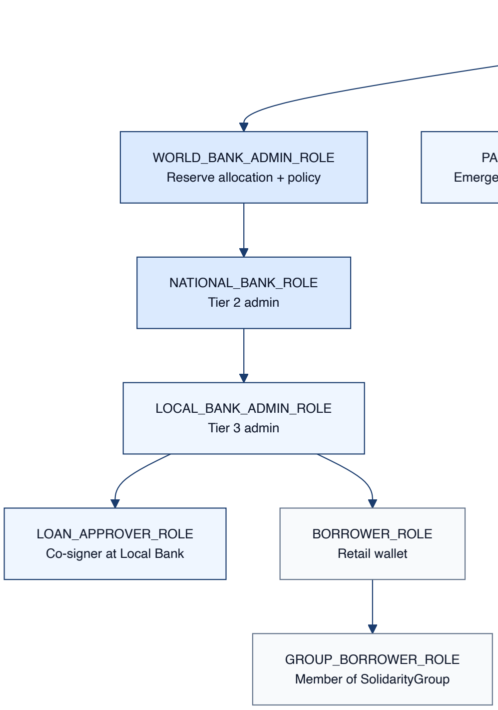
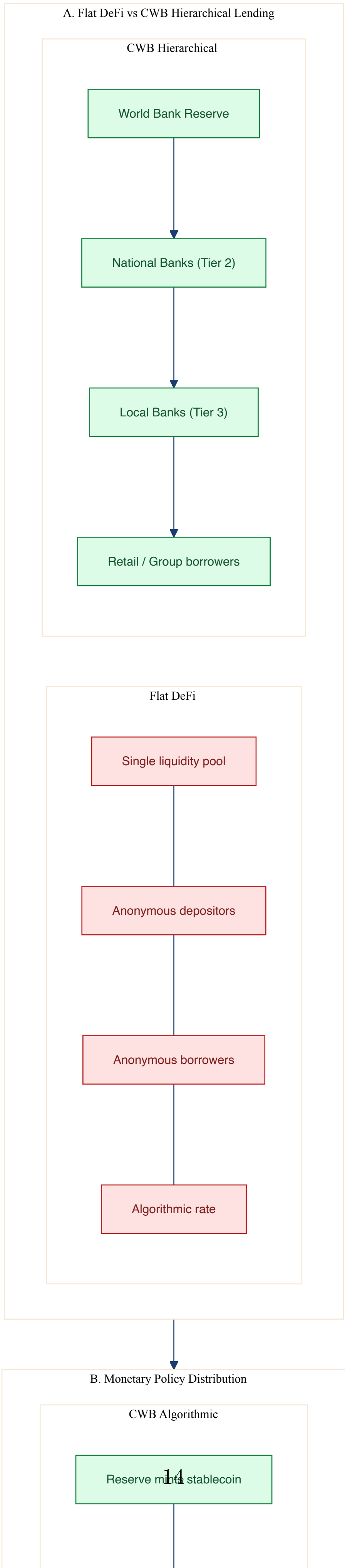



**Figure 1.5:** On-chain role hierarchy. DEFAULT_ADMIN_ROLE is held by a Safe multisig; downstream roles are scoped per contract.


A. Flat DeFi vs CWB Hierarchical Lending
CWB Hierarchical
World Bank Reserve
National Banks (Tier 2)
Local Banks (Tier 3)
Retail / Group borrowers
Flat DeFi
Single liquidity pool
Anonymous depositors
Anonymous borrowers
Algorithmic rate
B. Monetary Policy Distribution
CWB Algorithmic
Reserve mints stablecoin

# Chapter 2

# Literature Review

## 2.1 Preliminaries

This chapter situates Crypto World Bank within nine bodies of literature: (i) DeFi protocol design, (ii) interest-rate and liquidity mechanics of loanable funds, (iii) AI/ML and graph-neural-network fraud detection on blockchain, (iv) explainable AI in credit risk, (v) blockchain for financial inclusion and microfinance, (vi) smart-contract security and formal verification, (vii) zero-knowledge identity and AML compliance, (viii) federated learning for multi-institution risk intelligence, and (ix) LLM-augmented banking assistants. The review follows the PRISMA 2020 evidence-synthesis style adopted by recent blockchain-banking systematic reviews [pri(2025), flB(2024)].

### 2.1.1 Search protocol

The systematic search covered five databases – ACM Digital Library, IEEE Xplore, Elsevier ScienceDirect, MDPI, and Springer Link – with the time window 2020–2026, augmented by foundational pre-2020 works that remain canonical (Lundberg & Lee 2017 [Lundberg and Lee(2017)], Liu et al. 2008 [Liu et al.(2008)], Beck et al. 2018 [Beck et al.(2018)], Atzei et al. 2017 [Atzei et al.(2017)], Nakamoto 2008 [Nakamoto(2008)], Wood 2014 [Wood(2014)]). Inclusion criteria were peer-reviewed venue or established preprint, direct relevance to at least one CWB module, and an explicit methodological or empirical contribution rather than purely opinion-based reflection. Exclusion criteria removed retracted, single-blog-source, and venue-mismatched works. The result is a corpus of approximately 100 entries that are categorised below.

## 2.2 Review of Existing Research

### 2.2.1 DeFi protocol design and architecture

The Werner et al. *SoK: Decentralized Finance* [Werner et al.(2022)], published at ACM AFT 2022, is the definitive academic survey of the DeFi ecosystem; it maps the field into primitives, protocol types, and security categories and is the conceptual anchor for the four-tier hierarchical specification of CWB. Gudgeon et al. [Gudgeon et al.(2020)] formalise *Protocols for Loanable Funds* (PLFs) and derive utilisation-based interest-rate models that underpin formulas F.1, F.8, and F.16. Werner and Gudgeon both note that no surveyed protocol expresses an institutional hierarchy.

#### 2.2.2 Machine learning for blockchain security

Palaiokrassas et al. [Palaiokrassas et al.(2023)] apply ML to multichain DeFi fraud detection using behavioural features over raw transactions, and Hassan et al. [Hassan et al.(2022)] build a privacy-preserving inter-institutional ML pipeline on blockchain. The IEEE 2024 paper on real-time fraud detection in financial services [aib(2024)] embeds gradient boosting and RNNs directly inside permissioned smart contracts; throughput up to 1200 TPS and accuracy above 98%. Two contributions newly added in v14 are Liu et al. 2008 [Liu et al.(2008)] on Isolation Forest, the basis for the unsupervised anomaly channel in CWB's composite score (F.18), and the recent GNN-on-DeFi-fraud literature (Cheng et al. 2024, DeFiGuard, Wang & Wang 2025) [Cheng et al.(2024), Wang et al.(2024), Wang and Wang(2025)] which the CWB methodology will use as the ablation target for the future GNN upgrade.

#### 2.2.3 Explainable AI in financial decision systems

Lundberg & Lee [Lundberg and Lee(2017)] introduce SHAP, the basis for formula F.17 and for CWB's borrower-facing rejection explanations. Adom et al. [Adom et al.(2022)] compare LIME and SHAP on loan approval; the MDPI Risks 2024 paper [xai(2024)] demonstrates a SHAP-based XAI framework for credit risk and frames XAI explicitly as a regulatory tool. Bracke et al. [Bracke et al.(2019)] provide the Bank of England's analysis on ML explainability in default-risk modelling. Gupta et al. [Gupta et al.(2024)] demonstrate SHAP-based interpretable credit assessment.

#### 2.2.4 Blockchain for financial inclusion and microcredit

Mhlanga [Mhlanga(2022)] argues that blockchain reduces both the cost and the documentation friction that exclude rural borrowers from formal credit. Alam et al. [Alam et al.(2021)] pilot blockchain microcredit in Bangladesh. Howlader & Halder [Howlader and Halder(2025)] document the impact of mobile financial services on Bangladesh's inclusion gains. Kumar et al. [Kumar et al.(2023)] implement an Ethereum-based P2P lending framework specifically for SMEs and quantify the USD 5.2 trillion MSME credit gap. Toufaily & Zalan [Toufaily and Zalan(2024)] validate blockchain microcredit through industry interviews and identify coordination complexity and inter-institutional misalignment as the dominant barriers – exactly the problem the four-tier governance hierarchy addresses. Gücük et al. [Gücük et al.(2024)] survey privacy and trust in blockchain microlending.

#### 2.2.5 Smart contract security and formal verification

Atzei et al. [Atzei et al.(2017)] catalogue smart-contract attacks. Wang et al. [Wang et al.(2020)] (ContractWard), Liao et al. [Liao et al.(2019)] (SoliAudit), and So et al. [So et al.(2020)] (VeriSmart) compare automated vulnerability detection. The IEEE Network smart-contract vulnerability paper [sma(2020)] surveys reentrancy, integer overflow, and random-number defects with tool comparisons. Wu [Wu(2024)] addresses flash-loan vulnerability with static analysis. The arXiv 2025 exploitable-patterns study [exp(2025)] provides a recent vulnerability-to-defense taxonomy. The newest addition is the Certora 2025 open-source release [Certora(2025)] and the formal verification of Solidity contracts at industrial scale [cer(2023)] – the technical foundation for the CVL specification in Section 4.5.

#### 2.2.6 AI security features for blockchain lending

Mackinga et al. [Mackinga et al.(2023)] identify dynamic interest-rate-curve attacks. Albert et al. (GASOL) [Albert et al.(2020)] and Tolmach et al. [Tolmach et al.(2024)] address gas analysis and optimisation. The commit-reveal randomness paper [com(2025)] informs the AI risk oracle's commit-reveal design (Figure 3.3). Aponte-Novoa et al. [Aponte-Novoa et al.(2021)] examine 51% attacks.

#### 2.2.7 Identity, ZKP-KYC and AML compliance

Piper et al. [Piper et al.(2025)] (TU Berlin) propose privacy-preserving on-chain permissioning combining self-sovereign identity, verifiable credentials, and zk-SNARKs – the academic anchor for CWB's ZKP-KYC design. Decker [Decker(2025)] models ZKP for financial compliance broadly. Arshad et al. [Arshad et al.(2026)] demonstrate a Web3 identity and KYC framework at ACM TOW with 12.5-second average verification time and 40% compliance cost reduction. The newly added *zkAML* paper [zkAML Authors(2025)] extends ZKP to AML: proving non-membership in a sanctions Merkle root with 55 TPS throughput on public networks and 226 ms proof generation. Buterin et al. [Buterin et al.(2022)] establish the soulbound-token primitive, the basis for the CWB credit passport.

#### 2.2.8 Federated learning for multi-bank fraud intelligence

The *PrivChain-AI* paper (Nature Scientific Reports, 2025) [PrivChain-AI Authors(2025)] demonstrates federated learning over blockchain achieving 94.7% fraud detection accuracy without raw-data sharing. Abbassi et al. [Abbassi et al.(2025)] and *FED-SPFD* [FED-SPFD Authors(2025)] provide complementary federated frameworks specifically for financial fraud, all of which inform CWB's Section 4.3 federated specification.

#### 2.2.9 Gas cost optimization and Layer 2 scalability

Tolmach et al. [Tolmach et al.(2024)] optimise gas-fee structures; Albert et al. [Albert et al.(2020)] (GASOL) extract per-instruction gas costs from Solidity sources. These works underpin the storage-packing and event-indexing optimisations specified in Chapter 4.

### 2.2.10 Correspondent banking and cross-border settlement

The CPMI / BIS correspondent banking technical report [Committee on Payments and Market Infrast catalogues SWIFT correspondent chain inefficiencies; the FSB Stage 3 roadmap [Financial Stability Bo sets cross-border payments targets. Ripple's RLUSD analysis [Ripple(2026)] and JPMorgan's Kinexys [JPMorgan Chase(2025)] provide industry parallels.

### 2.2.11 Monetary policy distribution and financial inequality

Cantillon [Cantillon(1755)] is the historical origin of the eponymous effect. Beyer et al. [Beyer et al.(2025)] (2025) provide a current empirical analysis of the distributional impact of asset-purchase programmes. The Federal Reserve Board [Federal Reserve Board(2025)] provides US metropolitan-area evidence. The Human Rights Foundation [Human Rights Foundation(2025)] tracks CBDC privacy concerns. World Inequality Lab WIR 2026 [World Inequality Lab(2026)] provides the broader macro framing.

### 2.2.12 Real-world asset tokenization and institutional DeFi

GlobeNewswire / Corda RWA [GlobeNewswire(2025)], Centrifuge [Centrifuge(2025)], Maple Finance [Fensory(2026b)], Sky / MakerDAO [Fensory(2026c)], and DeFi credit platforms hitting USD 2.4 billion in loans [Fensory(2026a)] together establish that institutional DeFi has reached a credible inflection. The Sygnum 2026 institutional DeFi gap analysis [Sygnum Bank(2026)] is the explicit positioning anchor for CWB.

### 2.2.13 LLMs in finance

Nie et al. [Nie et al.(2023)] survey LLMs in finance at ACM ICAIF 2023 and provide the decision framework – prompt engineering, RAG, fine-tuning – that justifies CWB's QLoRA approach. LTIMindtree [LTIMindtree(2025)] and Aveni [Aveni(2025)] provide industry playbooks for fine-tuning financial LLMs. The 2025 adversarial-vulnerability paper [llm(2025)] motivates the red-teaming subsection (Section 4.2.4).

## 2.3 Literature Review Summary

Tables 2.1 and 2.2 consolidate the foundational and recent literature into two booktabs tables of twelve rows each, replacing the five fragmented summary tables of v13. They are organised by theme rather than by chronology.

## 2.4 Comparative Protocol Analysis

Table 2.3 positions CWB against the most-cited DeFi and TradFi peers along seven dimensions. The key empirical observation is that CWB is the only entry that satisfies

**Table 2.1:** Literature review summary, Part A: Foundations and architecture (12 entries).

| Ref. | Authors | Focus | Mapped CWB module | Year |
|---|---|---|---|---|
| [Werner et al.(2022)] | Werner et al. | SoK on DeFi taxonomy | Architecture / all layers | 2022 |
| [Gudgeon et al.(2020)] | Gudgeon et al. | Protocols for Loanable Funds | Interest & reserve (F.1, F.8) | 2020 |
| [Nakamoto(2008)] | Nakamoto | Bitcoin whitepaper | Foundational | 2008 |
| [Wood(2014)] | Wood | Ethereum Yellow Paper / EVM | Foundational | 2014 |
| [Aave(2022)] | Aave team | V3 technical paper | Lending engine pattern | 2022 |
| [Atzei et al.(2017)] | Atzei et al. | Attacks on Ethereum smart contracts | Security (Ch. 3.16) | 2017 |
| [Beck et al.(2018)] | Beck et al. | Blockchain governance framework | Governance (Ch. 3.15) | 2018 |
| [Buterin et al.(2022)] | Buterin et al. | Soulbound tokens (DeSoc) | Credit passport (3.6.4) | 2022 |
| [Lundberg and Lee(2017)] | Lundberg & Lee | SHAP unified explanation | XAI / F.17 | 2017 |
| [Liu et al.(2008)] | Liu et al. | Isolation Forest | Anomaly / F.14 | 2008 |
| [Piper et al.(2025)] | Piper et al. | On-chain ZKP-KYC permissioning | Identity (3.6.1) | 2025 |
| [Committee on Payments and Market Infrastructures(2016)] | CPMI / BIS | Correspondent banking analysis | Settlement comparison | 2016 |

the conjunction of *multi-tier hierarchy*, *AI-loan integration*, *ZKP-KYC and AML*, and *portable credit passport*.

### 2.4.1 Literature synthesis and research gaps

Across the bodies of literature surveyed, four explicit research gaps converge with the CWB design:

1. **Gap 1 – Hierarchical lending governance.** No flat DeFi protocol expresses an institutional tier system; TradFi institutions do but are not programmable. CWB occupies the previously empty multi-tier institutional quadrant (Figure 5.2).

2. **Gap 2 – AI-loan integration.** Existing AI-on-DeFi work is largely standalone (Random Forests in notebooks). CWB wires AI risk through a commit-reveal oracle into the lending workflow (Figure 3.3).

3. **Gap 3 – ZKP-AML.** ZKP work in DeFi addresses identity; sanctions and watchlist compliance is the open problem. CWB pairs KYC and AML circuits behind a single compliance gateway (Figure 3.6).

4. **Gap 4 – Portable credit history.** Repayment data is locked inside each protocol. CWB's CreditPassportSBT exposes a non-transferable, queryable record that any compatible lender can consult (Figure 3.8).

**Table 2.2:** Literature review summary, Part B: Recent applied work and new v14 additions (12 entries).

| Ref. | Authors / Venue | Focus | Mapped CWB module | Year |
|---|---|---|---|---|
| [Kumar et al.(2023)] | Kumar et al., ACM ICEA | Blockchain P2P SME lending | Group / SME (3.7.3) | 2023 |
| [Arshad et al.(2026)] | Arshad et al., ACM TOW | Web3 ZKP-KYC for Fin-Tech | Identity (3.6) | 2026 |
| [xai(2024)] | MDPI Risks | XAI for credit risk (SHAP) | XAI dashboard (4.2) | 2024 |
| [aib(2024)] | IEEE 2024 | AI + blockchain real-time fraud | ML pipeline (4.2) | 2024 |
| [Hassan et al.(2022)] | Hassan et al., IEEE Access | ML + blockchain fraud + privacy | Multi-bank FL (4.3) | 2022 |
| [Nie et al.(2023)] | Nie et al., ACM ICAIF | LLMs in finance survey | AI assistant (4.2) | 2023 |
| [sma(2020)] | IEEE Network | Vulnerability + audit tools | Security audit (3.16) | 2020 |
| [Toufaily and Zalan(2024)] | TFSC Elsevier | Blockchain microcredit (empirical) | Microfinance grounding | 2024 |
| [Cheng et al.(2024)] | Cheng et al., arXiv | GNN on DeFi fraud | GNN extension (4.4) | 2024 |
| [zkAML Authors(2025)] | IACR ePrint 2025/465 | zkAML circuit, 55 TPS | AML circuit (3.6.2) | 2025 |
| [PrivChain-AI Authors(2025)] | Nature Sci. Reports | PrivChain-AI FL (94.7%) | Federated (4.3) | 2025 |
| [Certora(2025)] | Certora OSS release | Certora Prover open source | Formal verification (4.6) | 2025 |

**Table 2.3:** Comparative protocol analysis. ✓ = supported, ▷ = partial, ○ = on roadmap, × = absent.

| Protocol | Multi-tier | AI-loan | ZKP-KYC | ZKP-AML | Credit SBT | Inst. governance |
|---|---|---|---|---|---|---|
| Aave V3 | × | × | × | × | × | ▷ |
| Compound V3 | × | × | × | × | × | ▷ |
| Morpho | × | × | × | × | × | ▷ |
| Maple Finance | ▷ | × | ▷ | × | × | ✓ |
| Goldfinch | ▷ | × | ▷ | × | × | ✓ |
| JPM Onyx | ✓ | ▷ | ✓ | ✓ | × | ✓ |
| BIS mBridge | ✓ | × | ✓ | ✓ | × | ✓ |
| Grameen / BRAC | ✓(off-chain) | × | × | × | × | ✓ |
| **CWB (this work)** | ✓ | ○ | ○ | ○ | ○ | ✓ |

## 2.5 Summary of Key Findings

The reviewed literature establishes that (i) DeFi has mature lending primitives but no institutional hierarchy; (ii) AI/ML on blockchain has mature individual techniques but no integrated risk pipeline; (iii) ZKP-based identity is now production-ready but AML and credit-history portability are not; (iv) institutional adoption of DeFi is blocked by governance and compliance gaps that an explicit hierarchical architecture can address. These four findings frame the system architecture in Chapter 3.

**Table 3.1:** Prototype scope: feature implementation status.

| Feature | Status |
|---|---|
| WorldBankReserve allocation & reserve invariants | ✓ |
| NationalBankContract scaffold (registration + allocation) | ▷ |
| LocalBankContract scaffold (registration + disbursement) | ▷ |
| LoanController (commit–reveal AI risk oracle wiring) | ◦ |
| LiquidationEngine (Health Factor enforcement) | ◦ |
| SavingsVault + FixedDeposit modules | ◦ |
| KYCVerifier (ZKP-KYC Circom 2.0 circuit) | ◦ |
| AMLVerifier (zkAML sanctions Merkle-root circuit) | ◦ |
| CreditPassportSBT (soulbound credit token) | ◦ |
| CWBTimeLock + Safe 3-of-5 multisig governance | ◦ |
| CWBPriceFeed (Chainlink adapter + staleness check) | ◦ |
| ERC-4337 Account Abstraction onboarding (Biconomy) | ◦ |
| The Graph subgraph for events | ◦ |
| FastAPI ML inference (RF + IF + SHAP) | ▷ |
| 9B QLoRA assistant (llama.cpp + ChromaDB RAG) | ◦ |
| Federated learning across Local Banks | ◦ |
| Certora CVL invariant verification (reserve contract) | ◦ |
| Foundry invariant + fuzz suite | ◦ |
| Tenderly runtime monitoring (6 alert triggers) | ◦ |
| Cross-chain bridge (LayerZero / CCIP candidate eval) | ◦ |

# Chapter 3

# System Architecture and Design

## 3.1 Prototype Scope

The pre-thesis 1 deliverable is an **architecture and specification document** together with a partial scaffold deployed on the Polygon Amoy testnet. The platform is explicitly not a production system: no real money is moved, no production KYC documents are processed, and operations are confined to the BRAC University internal pilot. Table 3.1 fixes the implementation status for every component referenced in this chapter; only items marked ✓ or ▷ are present in the current testnet build, while ◦ items have a specification in this chapter and are deferred to pre-thesis 2.

## 3.2 High-Level Architecture

### 3.2.1 Defense-in-depth architecture

The architecture is organised around a five-layer defense-in-depth model (Figure 3.1). Layer 1 is the smart-contract substrate: UUPS proxies, AccessControl with role-expiry timestamps, reentrancy guards, granular pause controls, KYC and AML on-chain gates, and a four-tool audit pipeline (Slither, Mythril, Echidna, Halmos / Certora). Layer 2 is the application layer: EIP-712 wallet authentication with 15-minute JWT, slowapi rate-limiting, CORS allowlists, Pydantic input validation, and a Redis-backed JWT blacklist. Layer 3 is the AI/ML defense: Random Forest fraud + Isolation Forest anomaly + SHAP top-3 explanations + behavioural session signals + federated multi-bank intelligence. Layer 4 is runtime monitoring: Tenderly alerts on six triggers, OpenZeppelin Defender automation, The Graph subgraph, and structured logs with correlation IDs. Layer 5 is operational: Safe multisig (2-of-3 demo, 3-of-5 production), hardware wallets for governance signers, and a key-rotation policy across demo, dev, and emergency-guardian keys.

### 3.2.2 Component view

Figure 3.2 expands the stack of Figure 1.3 into a component-and-interface view. The presentation layer is a React + Tailwind dApp with ethers.js, wagmi, and an Account-Abstraction SmartAccount SDK; the application layer is Express.js for orchestration, Socket.io for WebSocket dashboards, EIP-712 + JWT for authentication, slowapi for rate limiting, and a dedicated oracle relay service; the ML / LLM layer is FastAPI plus Random Forest, Isolation Forest, SHAP, the 9B QLoRA assistant, and ChromaDB RAG; the data layer combines PostgreSQL, Redis, S3 / MinIO, and a The Graph subgraph; the smart-contract layer hosts the reserve, two bank tiers, the liquidation engine, the SavingsVault, the KYCVerifier, the TimeLock, and the CreditPassportSBT; external services are Chainlink (Functions + Price Feeds), Firebase Auth, a KYC provider (Veriff or Persona), Tenderly, and Safe multisig.

## 3.3 Blockchain Platform Selection

### 3.3.1 Stablecoin-first loan denomination

Loans denominated in volatile native ETH expose retail borrowers to a doubling of real debt in a 50% drawdown – as occurred in May 2021 and 2022 [sta(2025)]. The CWB architecture therefore denominates retail-tier loans in USDC or USDT and elevates this from future work to a *design requirement*. The regulatory anchor is twofold: the EU MiCA framework (fully in effect December 2024) classifies stablecoins as E-Money Tokens requiring 1:1 reserve backing, authorisation, and regular audit [European Union(2024), Scorechain(2025)]; and the U.S. GENIUS Act (July 2025)

**Table 3.2:** Abbreviated MiCA and GENIUS Act compliance mapping (full mapping in Appendix E).

| Regulatory rule | CWB design choice | Status |
|---|---|---|
| MiCA: 1:1 reserve backing for E-Money Tokens | InsuranceFund + reserve-ratio enforcement on WorldBankReserve | ○ |
| MiCA: Authorisation before public offering | Sandbox-first pilot via Bangladesh Bank RFFO | ○ |
| MiCA: Regular audit | Certora CVL invariants + Tenderly continuous monitoring | ○ |
| GENIUS Act: Federal payment stablecoin framework | USDC integration, custody via licensed issuer | ○ |

creates the first federal payment-stablecoin framework [United States Congress(2025)]. Table 3.2 provides the abbreviated compliance mapping; the full mapping is in Appendix E.

### 3.3.2 Transaction verification and consensus

Polygon Amoy uses Proof-of-Stake with ∼5-second finality. Validator slashing, MEV-aware ordering, and the EIP-1559 fee market are inherited from the Polygon protocol. The block time and gas cost characteristics enable installment-payment loops at retail scale that would be uneconomic on mainnet Ethereum [Tolmach et al.(2024)].

### 3.3.3 Oracle architecture: AI to on-chain decision

On-chain contracts cannot execute Random Forests or call SHAP. A commit–reveal oracle (Figure 3.3) provides a manipulation-resistant path from off-chain ML to an on-chain loan decision: the off-chain service hashes the predicted score with a salt and commits the hash on-chain, then reveals the score after a one-block delay. The reveal window prevents the relay operator from re-running the model with knowledge of subsequent block data. The production path is Chainlink Functions, which replaces the centralised relay with a decentralised oracle network (Figure 3.4).

## 3.4 Data Model and Database Design

CWB persists nine core entities in PostgreSQL alongside on-chain state. Figure 3.5 is the consolidated ER diagram covering users, KYC credentials, banks, loans, repayments, AI/ML security logs, role changes, session events, liquidations, and credit-passport tokens. Five entities are new in v14 relative to v13: `kyc_credentials`, `session_events`, `role_changes`, `liquidations`, and `credit_passport_sbt`.

**Table 3.3:** Database entity summary. Five entities marked NEW in v14 support the new architecture sections.

| Entity | Purpose |
|---|---|
| `users` | Account, AA wallet, KYC level, guardian, role expiry |
| `kyc_credentials` | *(NEW)* per-level document hash, ZKP proof hash, expiry, revocation |
| `session_events` | *(NEW)* behavioural biometric events feeding Isolation Forest |
| `role_changes` | *(NEW)* off-chain mirror of on-chain RoleGranted / Revoked events |
| `banks` | Institutional registry: multisig address, on-chain hash, parent bank, tier |
| `loans` | Principal, term, interest, status, risk commitment tx, SHAP JSON |
| `repayments` | Installment ledger, transaction hash, payment time |
| `ai_ml_security_log` | Per-loan RF score, IF score, composite, SHAP top-3 |
| `liquidations` | *(NEW)* loanId, liquidator, bonus percentage, tx hash |
| `credit_passport_sbt` | *(NEW)* closed-loans, defaults, liquidations, score, token URI |
| `messages` | Borrower–approver chat threads |

**Table 3.4:** PostgreSQL index plan.

| Index | Query pattern | Notes |
|---|---|---|
| `idx_loans_borrower` | rolling-limit SQL by wallet | supports F.19 |
| `idx_loans_local_bank` | bank-operator dashboard | status filter pushed down |
| `idx_loans_created_at` | time-window scans | supports 180d/365d limits |
| `idx_repayments_loan` | installment view per loan | ordered by installment_no |
| `idx_aiml_wallet` | risk history per wallet | DESC ts (latest first) |
| `idx_bank_users_wallet` | role binding lookup | unique partial index on active |
| `idx_session_user_ts` | behavioural feature window | last 90 days inclusive |

### 3.4.1 Entity summary

### 3.4.2 Normalization

The schema is third-normal-form: every non-key attribute depends on the whole primary key. `ai_ml_security_log` is separated from `loans` because risk scores can be regenerated independently and audit history may exceed loan lifespan. `kyc_credentials` is separated from `users` because multiple credentials per user (level promotion, reissuance, revocation) must be retained for audit.

**Table 3.5:** On-chain / off-chain data classification matrix.

| Data item | Stored at | Rationale | Privacy class |
|---|---|---|---|
| Loan principal, term, status | On-chain (LocalBank) | Required by contract logic | Public (pseudonymous wallet) |
| Risk score commitment hash | On-chain (LoanController) | Tamper-evident audit | Public (hash only) |
| Repayment record | On-chain (LocalBank) | Required for HF and SBT update | Public |
| Health Factor inputs (C, D, LT) | On-chain | Required by LiquidationEngine | Public |
| Reserve ratio invariants | On-chain | Required by reserve contract | Public |
| Bank registration metadata | On-chain hash, off-chain doc in S3 | Privacy + integrity | Confidential |
| KYC level flag | On-chain (KYCVerifier) | Required by loan gate | Public (flag only) |
| KYC documents | S3 / MinIO encrypted at rest | Privacy + GDPR / PDPA | PII – restricted |
| ZKP proof of KYC / AML | On-chain (verifier accepts) | Required by gate | Public proof, private witness |
| Session events (biometric) | PostgreSQL | ML feature, no contract use | Pseudonymous |
| SHAP top-3 explanation | PostgreSQL + hash on-chain | Borrower-facing, auditable | Pseudonymous |
| Chat messages | PostgreSQL | No contract use | Confidential |
| Market data cache | Redis (TTL 60–300s) | Throughput | Public |

### 3.4.3 Indexing strategy

### 3.4.4 Functional dependencies and relational integrity

The principal FDs are: `users.wallet_address` → `users.id`; `loans.id` → (`borrower_id`, `local_bank_id`, `amount`, `status`); `repayments.id` → (`loan_id`, `installment_no`, `amount`, `paid_at`); `kyc_credentials.id` → (`user_id`, `level`, `document_hash`, `expires_at`). Foreign-key constraints use `ON DELETE RESTRICT` for financial relationships and `ON DELETE SET NULL` for audit-only relationships.

## 3.5 On-Chain and Off-Chain Data Partitioning

The decision rule is: *state required by smart contract logic, or required for tamper-evident audit, lives on-chain; everything else lives off-chain.* Table 3.5 fixes this for every data category in v14.

## 3.6 Digital Identity System

### 3.6.1 ZKP-KYC compliance architecture

The KYC circuit follows the design of Piper et al. [Piper et al.(2025)] and Arshad et al. [Arshad et al.(2026)]: a Circom 2.0 circuit proves (i) the user holds a Verifiable Credential signed by an approved issuer, (ii) the user's birth year places them above 18, and (iii) the user's country is in the allow-set for the requested operation, without revealing any of the underlying values. The proof is verified by an on-chain `KYCVerifier` contract; only the verified KYC level (0–3) is stored on-chain.

### 3.6.2 ZKP-AML against sanctions

The AML circuit, following the zkAML paper [zkAML Authors(2025)], proves that the user's wallet address is *not* a member of a sanctions Merkle root that is refreshed daily from OFAC, TRM Labs, and Chainalysis watchlists. The on-chain `AMLVerifier` verifies the proof and stores a clean flag. Both proofs together comprise the compliance gateway (Figure 3.6). Reported throughput on public networks is 55 transactions per second with 226 ms proof generation; this is within the latency budget for loan onboarding.

### 3.6.3 W3C Decentralized Identifiers and Verifiable Credentials

The architecture anchors identity through W3C Decentralized Identifiers (DIDs) backed by Verifiable Credentials (VCs). The DID is registered on-chain by the user's smart account; the KYC VC is issued by an external provider and held in the user's wallet, never on-chain. The ZKP circuit proves possession of a valid VC without exposing the DID or credential content (Figure 3.7).

### 3.6.4 Soulbound credit passport

The CreditPassportSBT is a non-transferable ERC-5114-like token [Buterin et al.(2022)] that records each borrower's repayment history. The token is minted on the first closed loan; updated on every loan closure, default, and liquidation; and queryable by any compatible lending protocol. Figure 3.8 captures the mint–update–attest lifecycle.

## 3.7 User Taxonomy and Onboarding

CWB recognises eleven distinct user types covering anonymous browsers, retail borrowers, group borrowers and leaders, bank operators and approvers at each tier, KYC providers, backup guardians, and Safe multisig signers. Onboarding is a five-stage funnel (Figure 3.9).

**Table 3.6:** Cross-tier transfer function matrix.

| Function | Caller role | Target |
|---|---|---|
| `allocateToNationalBank` | WORLD_BANK_ADMIN_ROLE | NationalBankContract |
| `allocateToLocalBank` | NATIONAL_BANK_ROLE | LocalBankContract |
| `disburseLoanToUser` | LOCAL_BANK_ADMIN + APPROVER | Borrower wallet |
| `returnSurplusToParent` | Any (LB or NB) when surplus > floor | Parent contract |
| `liquidate` | Any keeper | LiquidationEngine |
| `commitRiskScore` | ORACLE_ROLE | LoanController |
| `revealRiskScore` | ORACLE_ROLE (after 1-block delay) | LoanController |

### 3.7.1 Tiered KYC

The KYC ladder has four levels (Figure 3.10). Level 0 is anonymous read-only access. Level 1 adds email and phone verification and unlocks a 0.1 ETH equivalent limit. Level 2 adds NID + selfie liveness via Veriff or Persona and unlocks 2 ETH equivalent. Level 3 adds income proof and address verification, optionally upgraded to a ZKP-KYC proof, and unlocks 10 ETH equivalent.

### 3.7.2 Group solidarity formation

The group lending flow (Figure 3.11) implements the BRAC and Grameen solidarity model on-chain.

## 3.8 Kinked Interest Rate Model

The platform follows the kinked utilisation-rate model from Aave V3 [Aave(2022), Gudgeon et al.(2020)]: borrow APR rises slowly with utilisation $U$ (formula F.1) below a kink point (typically $U^* = 0.8$) and steeply above it. Two segments suffice for the prototype: $\mathrm{APR}(U) = R_0 + (R_1/U^*) \cdot U$ for $U \leq U^*$, and $\mathrm{APR}(U) = R_0 + R_1 + R_2 \cdot (U - U^*)/(1 - U^*)$ for $U > U^*$, with $R_0 = 3\%, R_1 = 2\%, R_2 = 15\%$ as initial parameters.

## 3.9 Cross-Tier Capital Flow

Three downward functions allocate capacity: `allocateToNationalBank`, `allocateToLocalBank`, and `disburseLoanToUser` (gated by the AI risk oracle). One upward function, `returnSurplusToParent`, repatriates surplus collected interest back up the hierarchy. The LiquidationEngine is wired sideways to the LocalBank. The complete topology is Figure 3.12.

## 3.10 Liquidation Engine

The LiquidationEngine implements the Health Factor formula (F.7) with on-chain enforcement. Figure 3.13 shows the keeper flow: any participant may call `checkLoan(loanId)`; if HF $< 1.0$ the keeper may call `liquidate(loanId)`, the engine seizes the collateral, transfers a 5–8% bonus to the keeper, and writes a $-\Delta$ event to the borrower's CreditPassportSBT [Aave(2022)].

## 3.11 SavingsVault and FixedDeposit Module

A bank is fundamentally a deposit institution. The SavingsVault is specified as a variable-APY pool tied to utilisation, with on-chain interest accrual and withdrawal gated by the platform reserve ratio (F.8). The FixedDeposit contract complements the vault with duration-matched liabilities that fund installment loans, addressing the asset-liability management dimension of formula F.16.

## 3.12 System Modeling

### 3.12.1 Use case diagram

Figure 3.14 shows the use-case map with thirteen actors and seventeen use cases, including include / extend relations for KYC, AML, and oracle integration.

### 3.12.2 Activity and sequence diagrams

The loan-request activity (Figure 3.15), the consolidated communication flows (Figure 3.16), the income-verification activity and sequence (Figure 3.17), and four planned-feature activities (Figure 3.18) cover the user-side workflows. Three sequence diagrams cover the loan approval / rejection (Figure 3.20, merged reject path), the installment loop (Figure 3.21), and the borrowing-limit calculation (Figure 3.22). The hierarchical banking flow is consolidated with its activity counterpart in Figure 3.23 and the market data flow in Figure 3.24.

## 3.13 Banking Product Suite

The product suite – savings, checking, group lending, FX, planned trade finance, planned privacy-preserving identity compliance – is the user-facing surface of the architecture. Each product maps to one or more contracts in Table 1.1.

## 3.14 Governance Framework

### 3.14.1 Safe multisig and TimeLock

Every governance action passes through a Safe multisig, then queues to a TimeLock with a 24–48 hour delay (5 minutes in the demo environment). Figures 3.25, 3.26, and 3.27 show the propose–queue–execute sequence, the UUPS upgrade path, and granular pause states.

**Table 3.7:** Consolidated governance framework (network membership, business, technology, regulatory).

| Layer | Mechanism | Enforcement |
|---|---|---|
| Network Membership | WB Admin grants NATIONAL_BANK_ROLE | On-chain RoleGranted + off-chain role_changes mirror |
| Business Network | Interest-rate parameters, reserve ratios, KYC requirements | DAO vote → Safe multisig → TimeLock → contract setter |
| Safe Multisig + TimeLock | 2-of-3 demo, 3-of-5 production; 24–48h delay | CWBTimeLock + Safe at top of role tree |
| Technology Infrastructure | UUPS proxy upgrades, Chainlink staleness, oracle role granting | UUPS `_authorizeUpgrade` restricted to UPGRADER_ROLE |
| Regulatory Compliance | MiCA + GENIUS reserves; Bangladesh sandbox path; over-indebtedness caps | Reserve invariants + sandbox license + cap on simultaneous loans |

**Table 3.8:** Threat model and primary mitigation.

| Threat | Layer | Primary mitigation |
|---|---|---|
| Reentrancy | L1 contract | OpenZeppelin ReentrancyGuard on disburse, repay, withdraw |
| Integer overflow | L1 contract | Solidity 0.8+ checked arithmetic |
| Oracle manipulation | L1 + L3 ML | Commit-reveal + Chainlink Price Feed staleness check |
| Unauthorised disburse | L1 contract | AccessControl role expiry + Foundry fuzz coverage |
| Flash-loan-driven liquidation [Wu(2024)] | L1 contract | Atomic HF re-check inside liquidate function |
| Bridge exploit | L1 cross-chain | Confine state to one chain pre-MVP; evaluate LayerZero / Axelar / CCIP |
| JWT theft / replay | L2 application | 15-minute access TTL + jti blacklist in Redis on logout |
| API abuse / scraping | L2 application | slowapi 5 / 60 / 10 per minute per IP |
| Blind signing of malicious tx | L2 application | Decode calldata in UI before display |
| KYC provider outage | L4 monitor | Allow Level-1 fallback + queue document for later |
| Safe signer compromise | L5 op sec | 3-of-5 threshold + hardware wallet + key rotation |
| Sanctions watchlist false negative | L1 + ZKP | Daily AML Merkle root refresh + manual override flag |

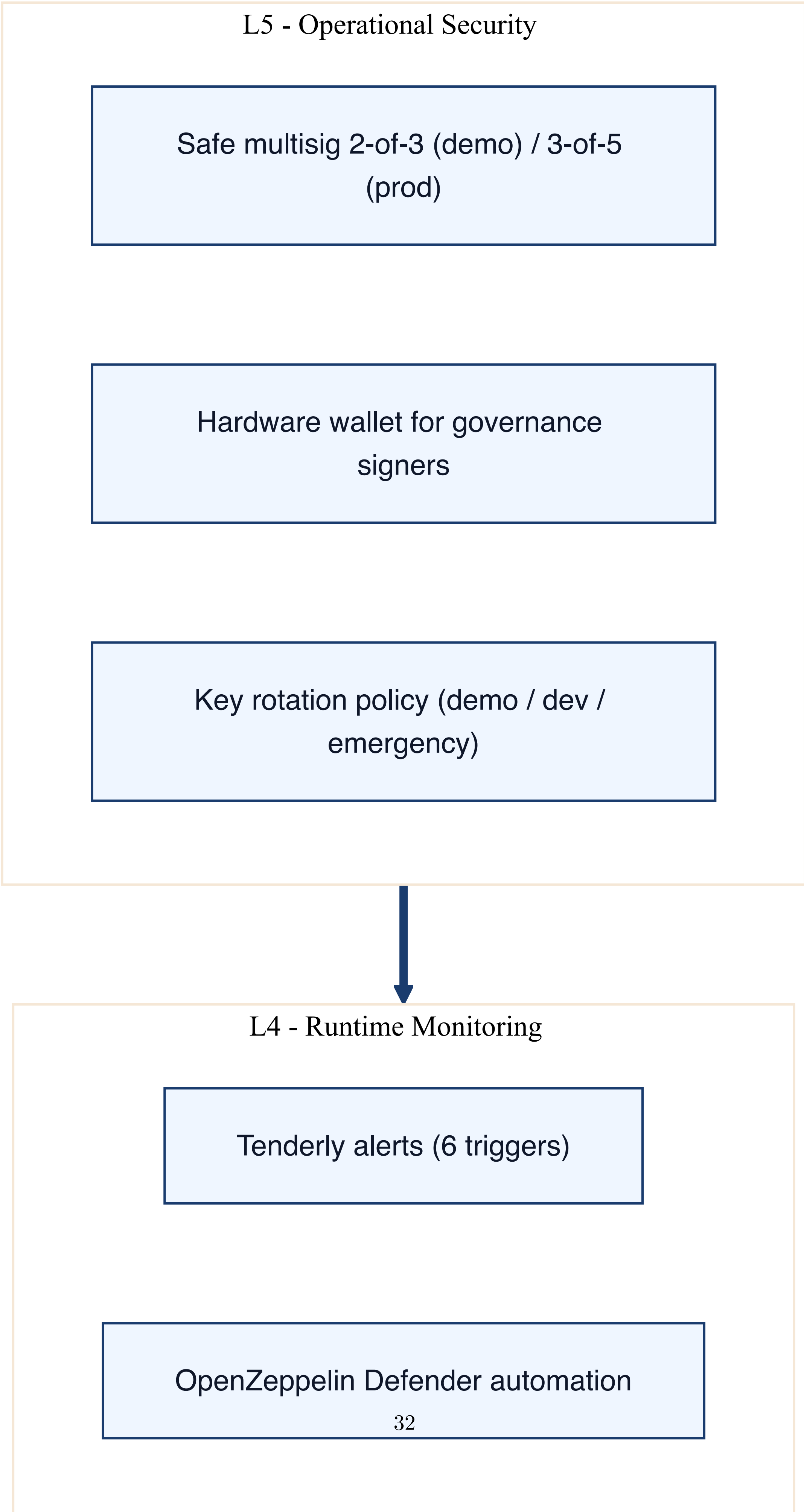
**Table 3.9:** SWC Registry mapping for the most security-critical functions.

| Function | SWC ID | Vulnerability class | Mitigation |
|---|---|---|---|
| `disburseLoanToUser` | SWC-105 | Unprotected ether withdrawal | Role + KYC + AML gates |
| `payInstallment` | SWC-107 | Reentrancy | ReentrancyGuard + checks-effects-interactions |
| `liquidate` | SWC-128 | DoS with block gas limit | Single loan per call; keepers paid bonus |
| `commitRiskScore` | SWC-114 | Transaction order dependence | Hash commitment + reveal delay |
| `upgradeTo` | SWC-112 | Delegatecall to untrusted callee | UPGRADER_ROLE + TimeLock |
| `withdraw` (SavingsVault) | SWC-107 | Reentrancy | ReentrancyGuard |
| Reserve ratio setter | SWC-115 | Authorization via `tx.origin` | Use `msg.sender` + role check |

## 3.15 Threat Model and Security Controls

## 3.16 Cross-Chain Bridge Architecture

Multi-chain deployment introduces cross-chain state-consistency risk: a borrower with an active loan on one chain must not be able to bypass tier limits by opening a second loan on another chain. The pre-thesis 2 deliverable will evaluate three candidate bridge protocols – LayerZero, Axelar, and Chainlink CCIP – and specify whether (a) loan state is replicated cross-chain, (b) loan state is confined to one chain per borrower (the CreditPassportSBT records the chain-of-origin), or (c) a canonical chain is the source of truth and other chains are read-replicas. Bridge hacks were the largest single category of crypto loss between 2022 and 2024; the architecture cannot leave this question open at production scale.


L5 - Operational Security
Safe multisig 2-of-3 (demo) / 3-of-5 (prod)
Hardware wallet for governance signers
Key rotation policy (demo / dev / emergency)
L4 - Runtime Monitoring
Tenderly alerts (6 triggers)
OpenZeppelin Defender automation

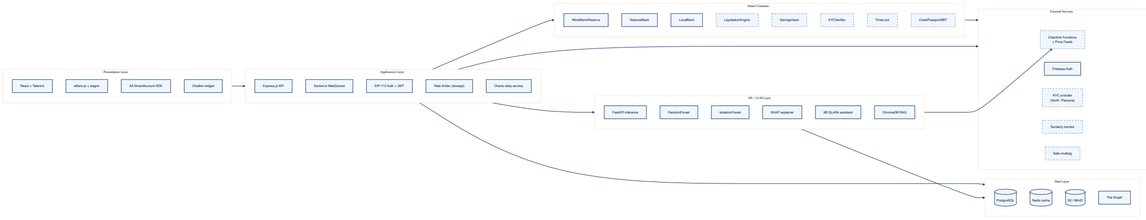



**Figure 3.2:** Component diagram. Solid components are implemented or scaffolded; dashed components are planned for pre-thesis 2.

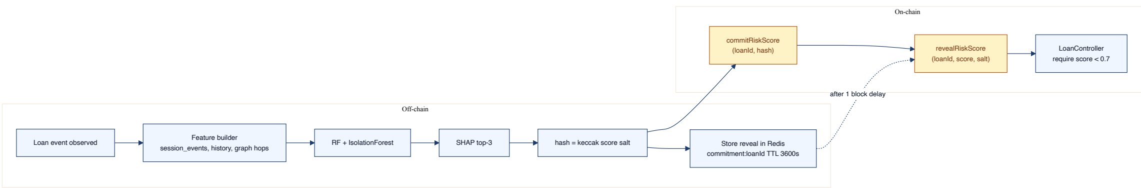



**Figure 3.3:** Commit–reveal oracle pipeline for AI-driven loan scoring.

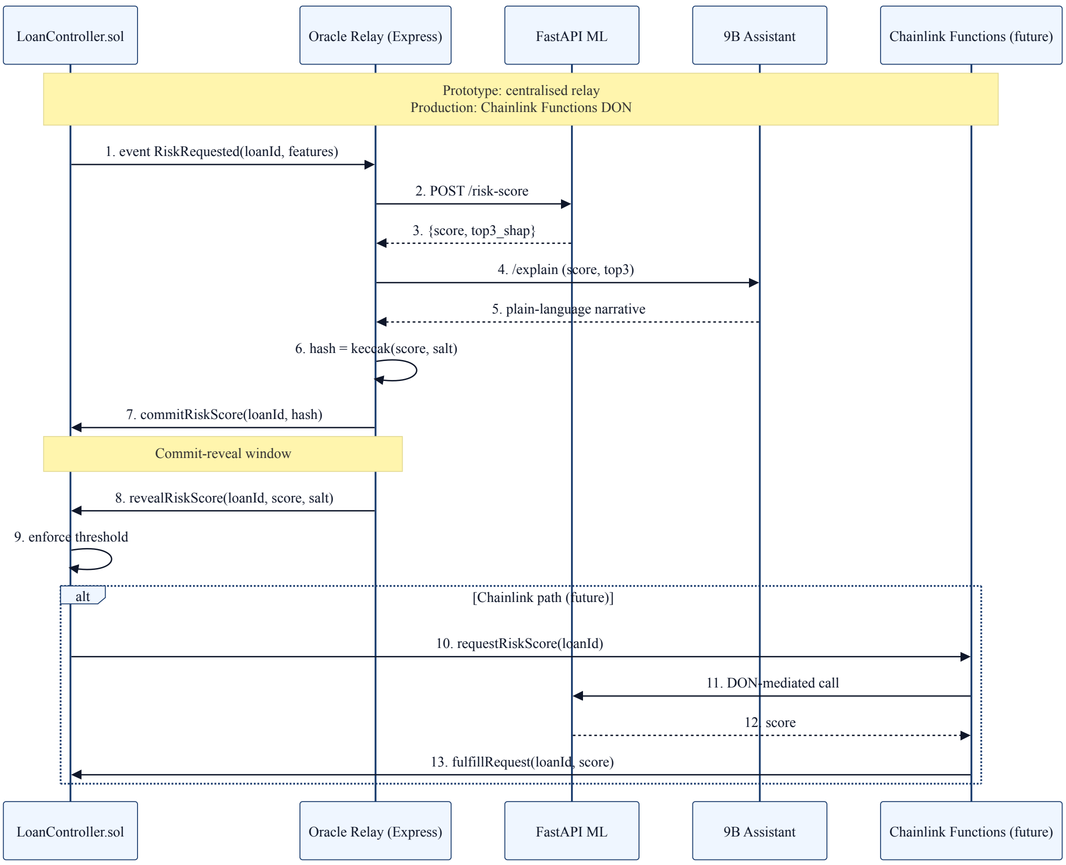



**Figure 3.4:** AI oracle integration: prototype uses centralised commit-reveal; production migrates to Chainlink Functions.

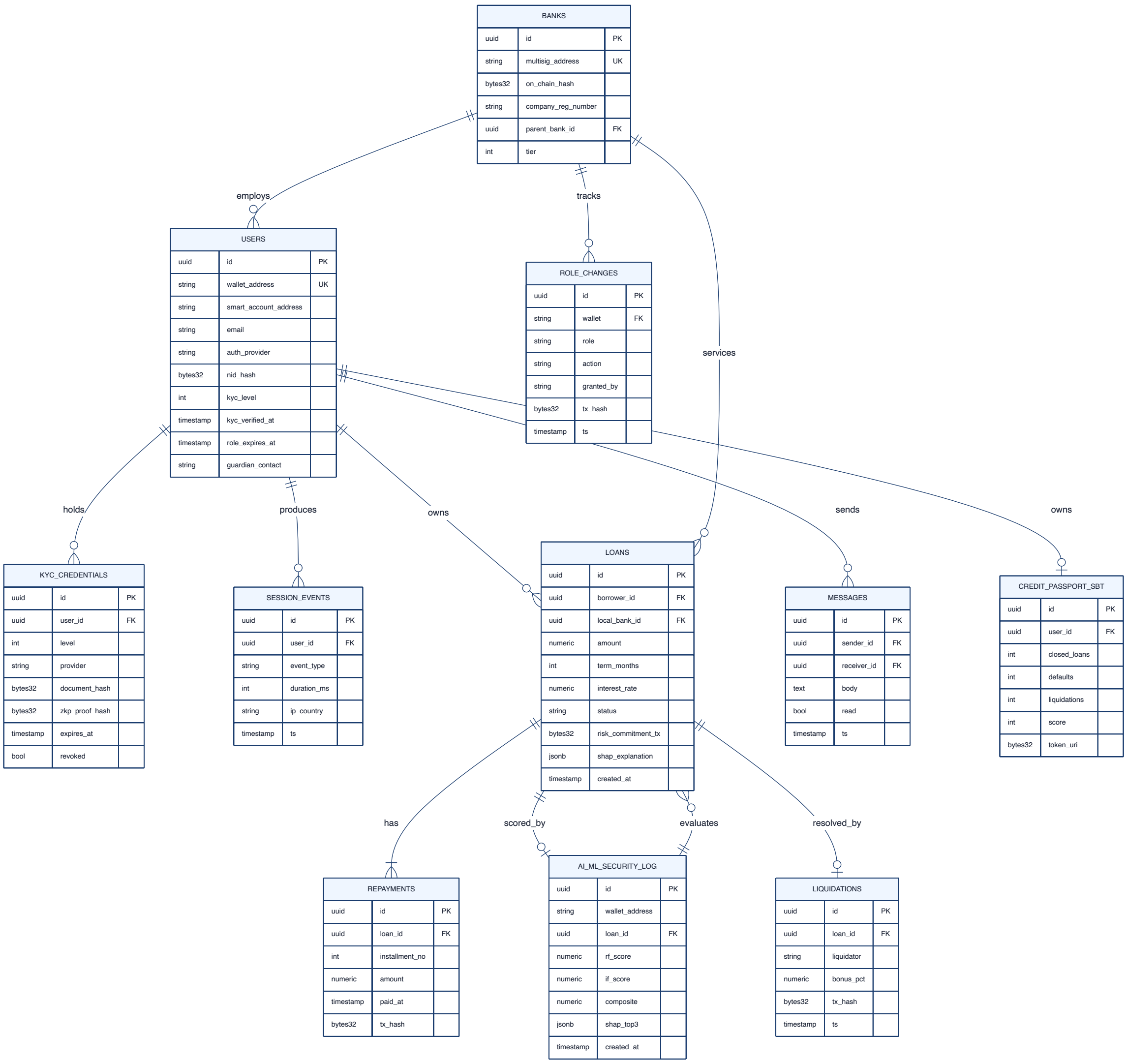



**Figure 3.5:** Consolidated CWB Entity Relationship Diagram.

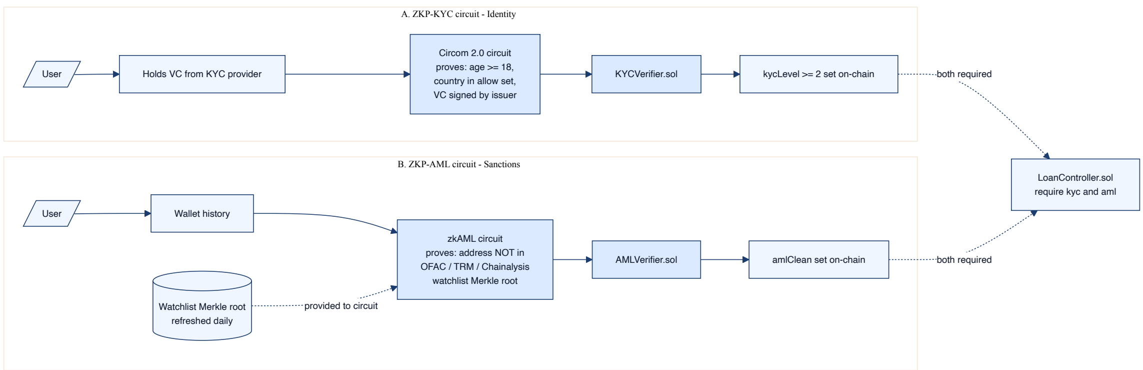



**Figure 3.6:** Dual ZKP compliance gateway: separate circuits for identity and sanctions.

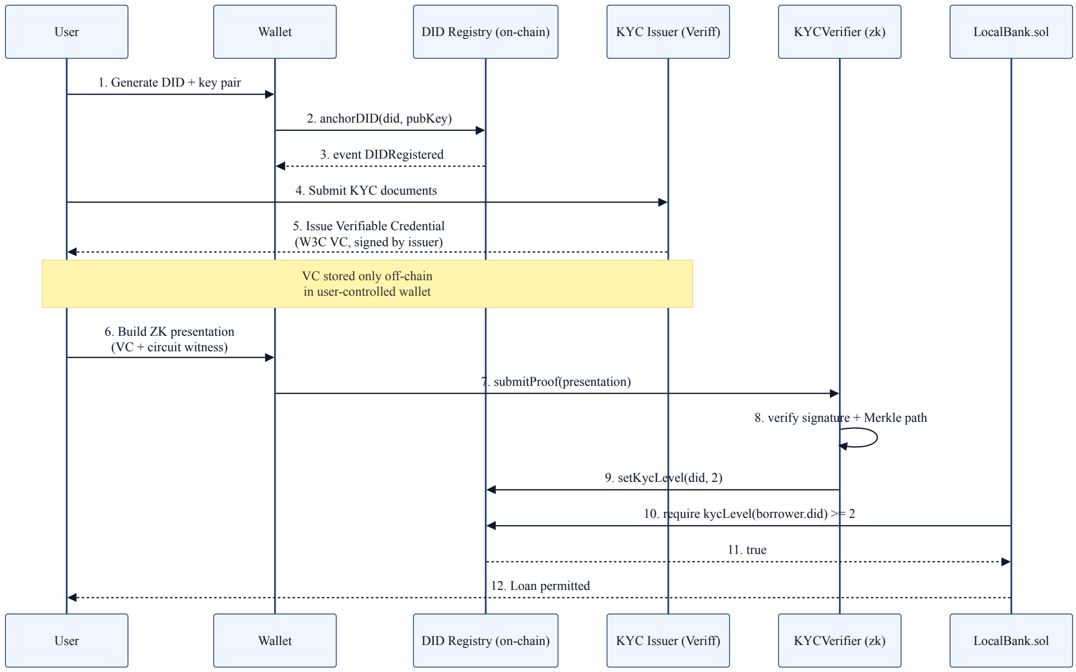



**Figure 3.7:** W3C DID and Verifiable Credential issuance plus ZKP presentation.

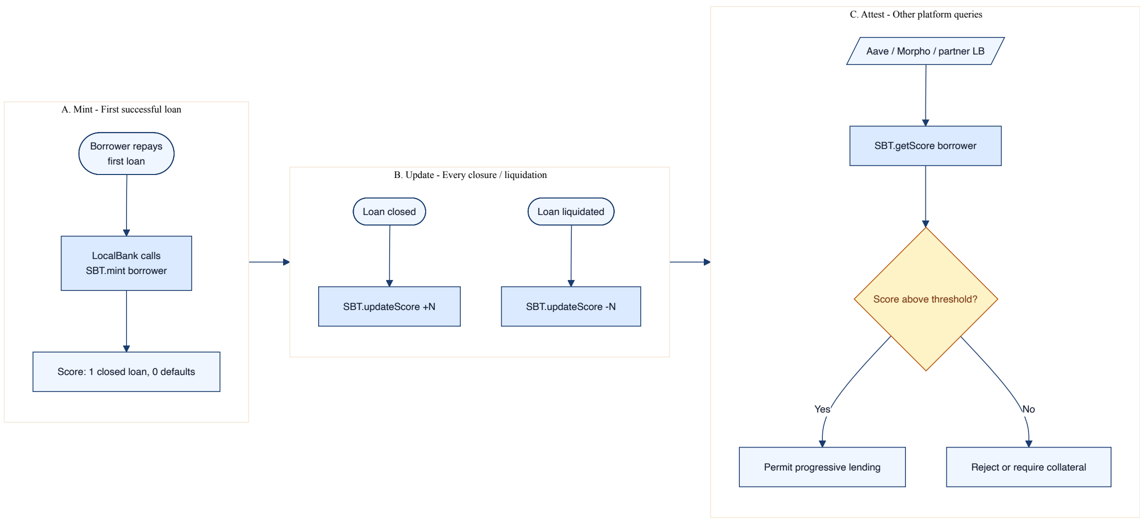



**Figure 3.8:** Soulbound credit passport lifecycle: mint, update, attest.

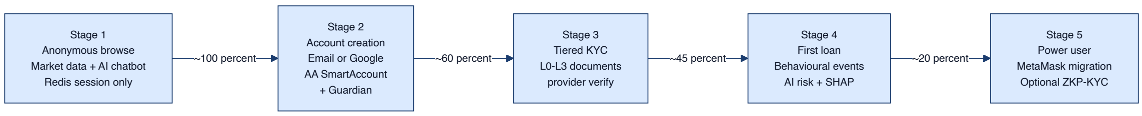



**Figure 3.9:** Five-stage onboarding funnel with abstracted-account retail entry and optional MetaMask migration at the power-user stage.

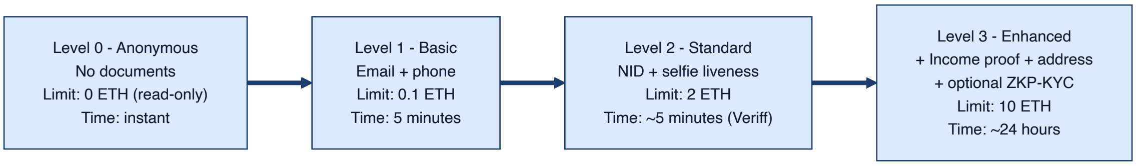



**Figure 3.10:** Four-level tiered KYC ladder.

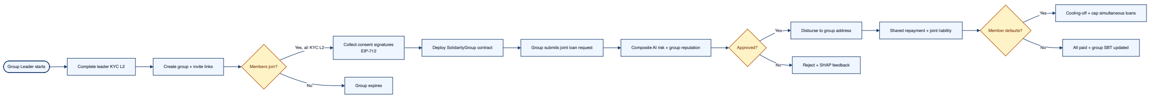



**Figure 3.11:** Group solidarity formation and joint-loan flow.

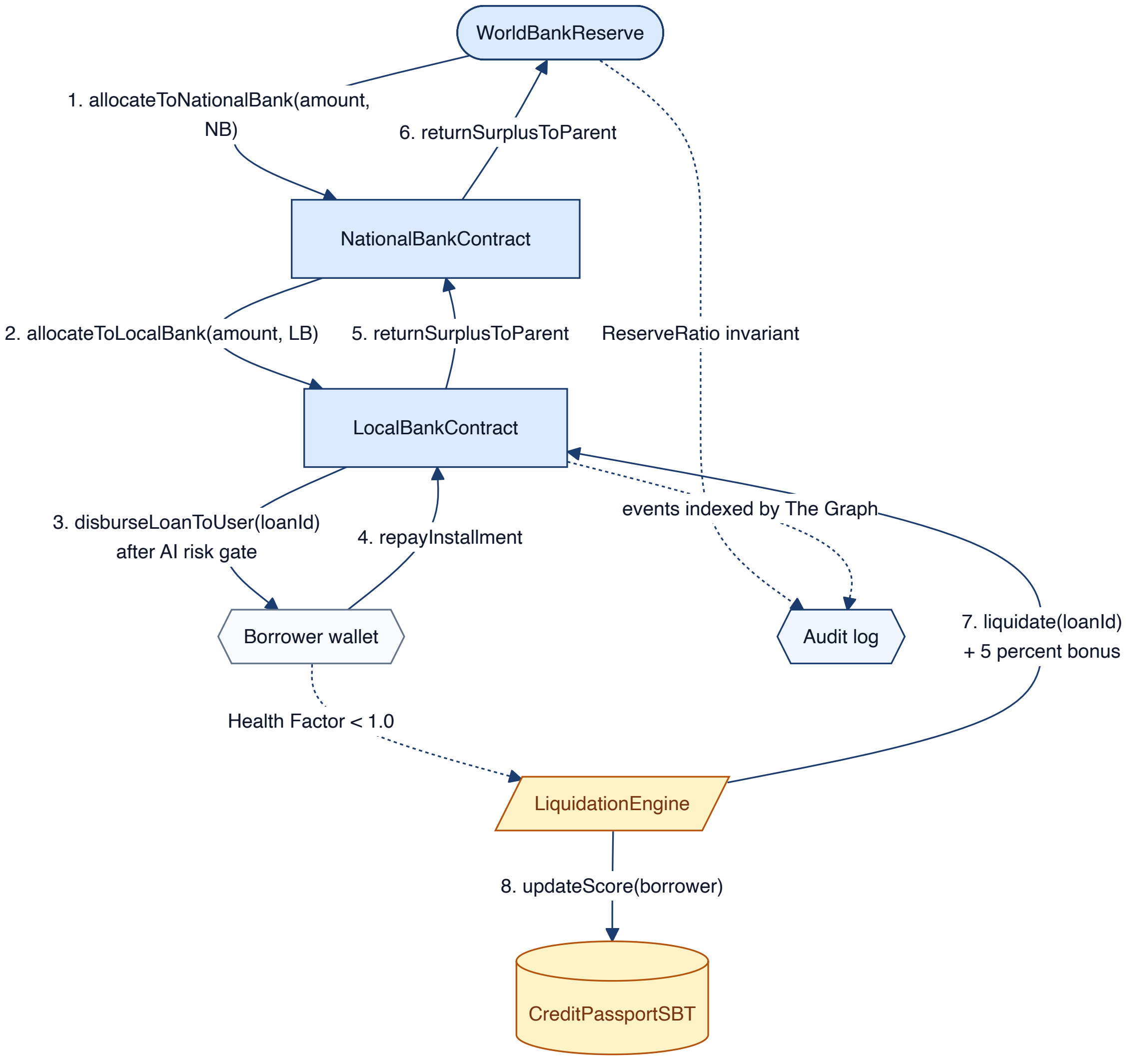



**Figure 3.12:** Cross-tier capital flow with downward allocation, upward surplus repatriation, and the liquidation hook.

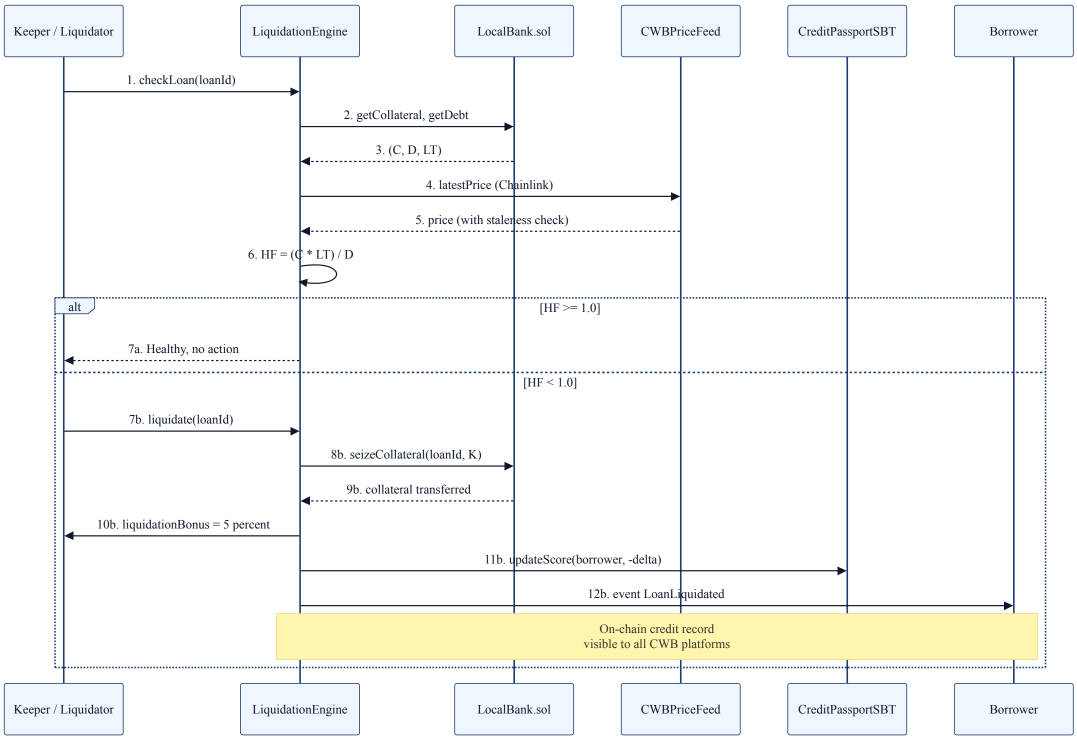
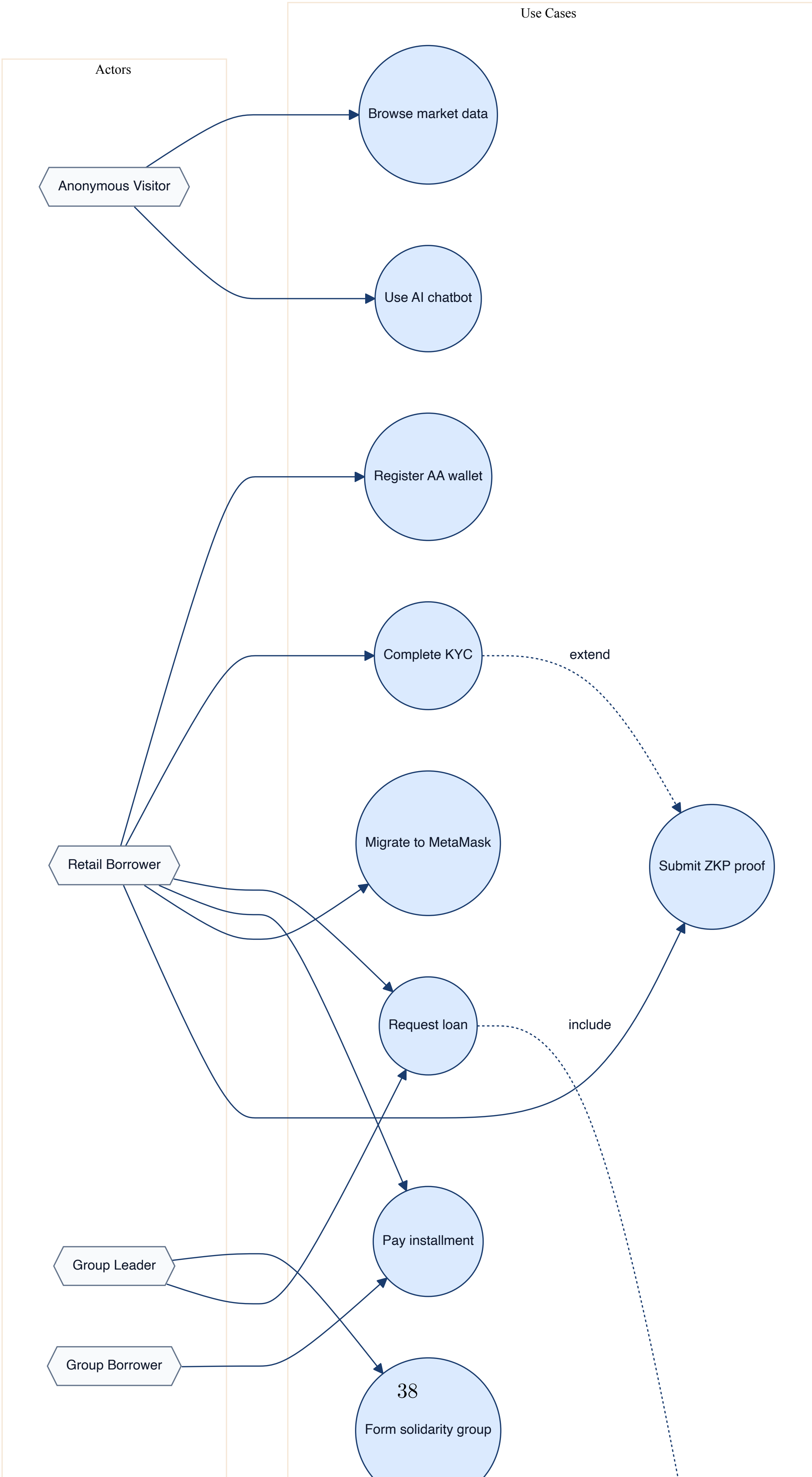



**Figure 3.13:** LiquidationEngine sequence: Health Factor check, liquidation call, keeper bonus, and SBT update.


Use Cases
Actors
Browse market data
Anonymous Visitor
Use AI chatbot
Register AA wallet
Complete KYC
extend
Migrate to MetaMask
Submit ZKP proof
Retail Borrower
Request loan
include
Pay installment
Group Leader
Group Borrower
Form solidarity group

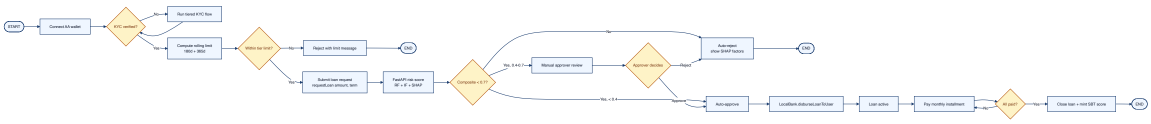



**Figure 3.15:** End-to-end borrower activity: connect AA wallet, KYC, rolling limit, AI risk scoring, approval, disbursement, installment loop, SBT update.

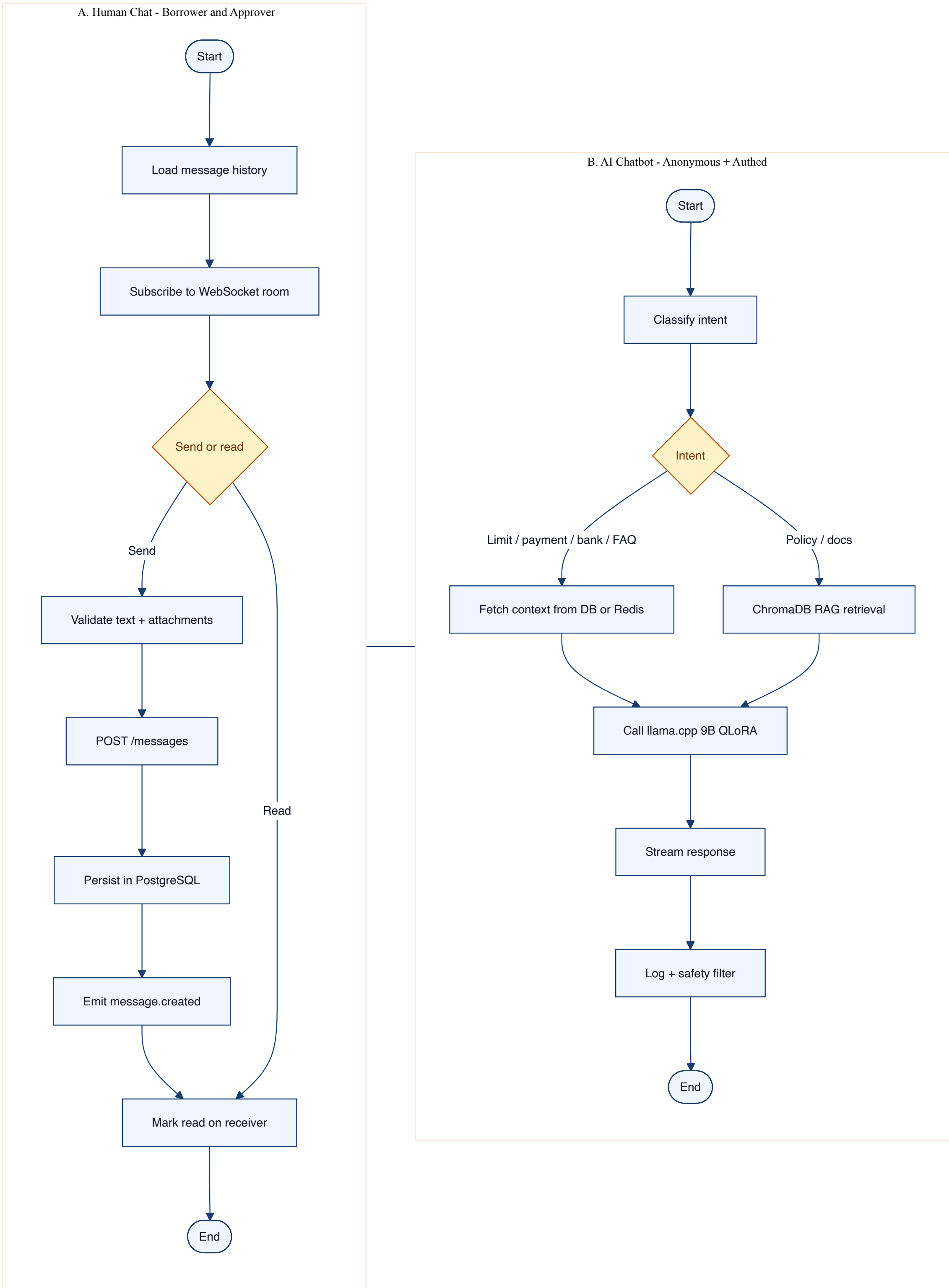



**Figure 3.16:** Consolidated communication activities. Panel A: human chat. Panel B: AI chatbot.

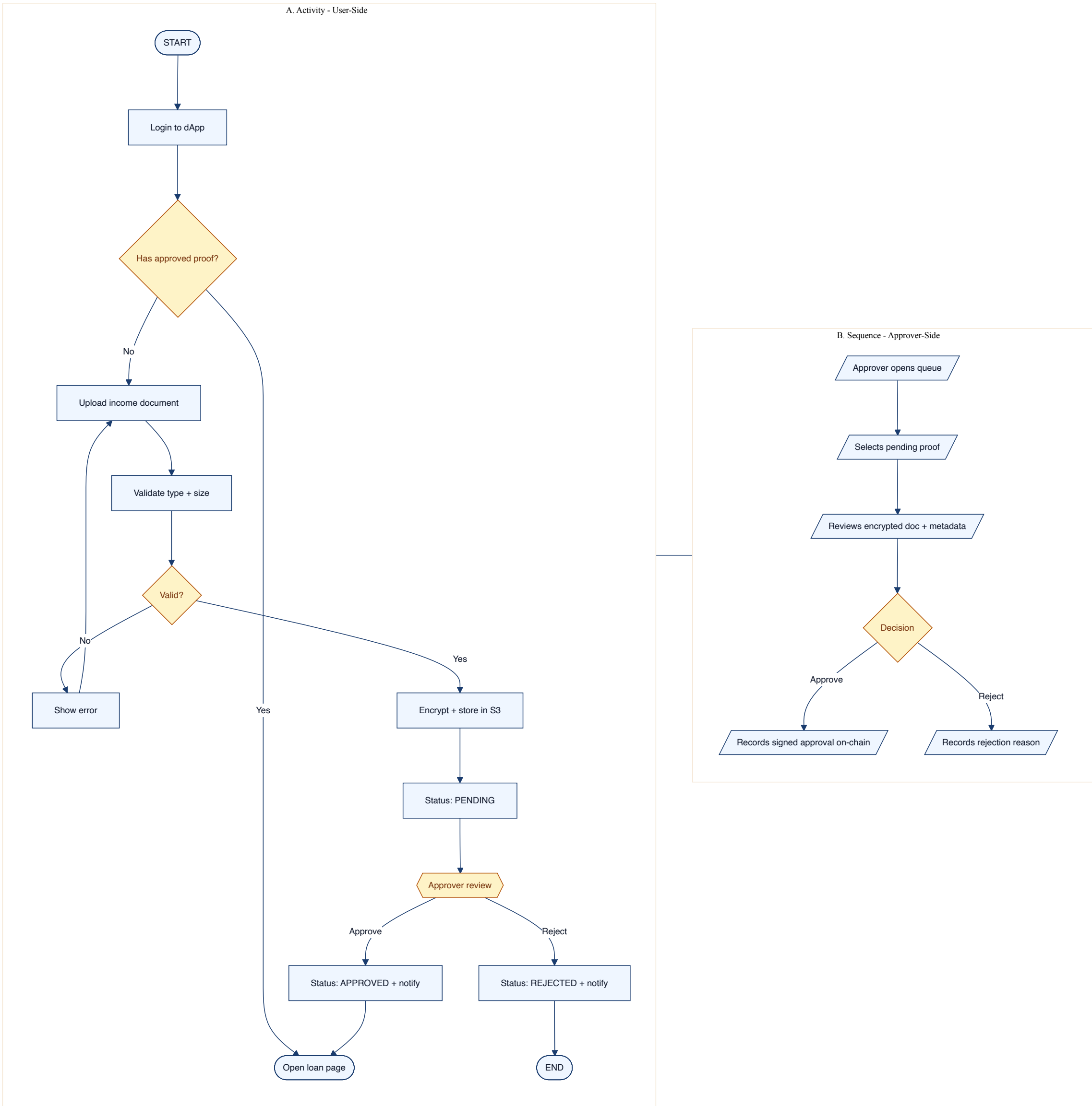



**Figure 3.17:** Income verification: user activity (Panel A) and approver sequence (Panel B).

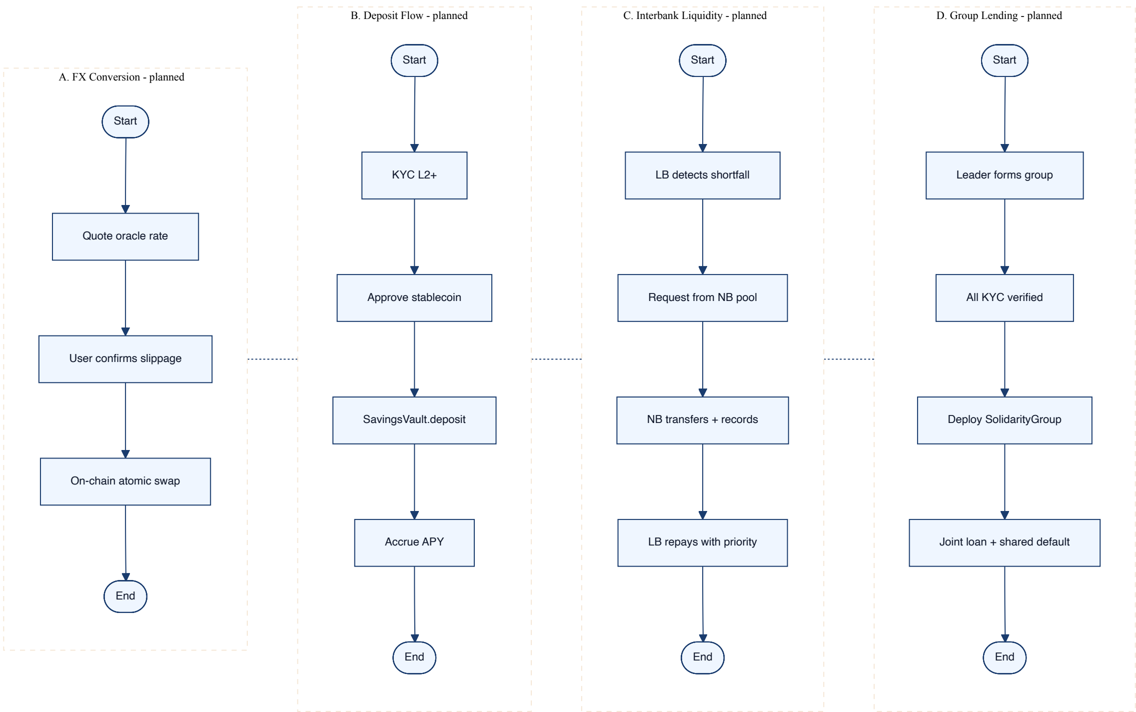



**Figure 3.18:** Four planned feature activities: FX conversion, deposit flow, interbank liquidity, group lending.

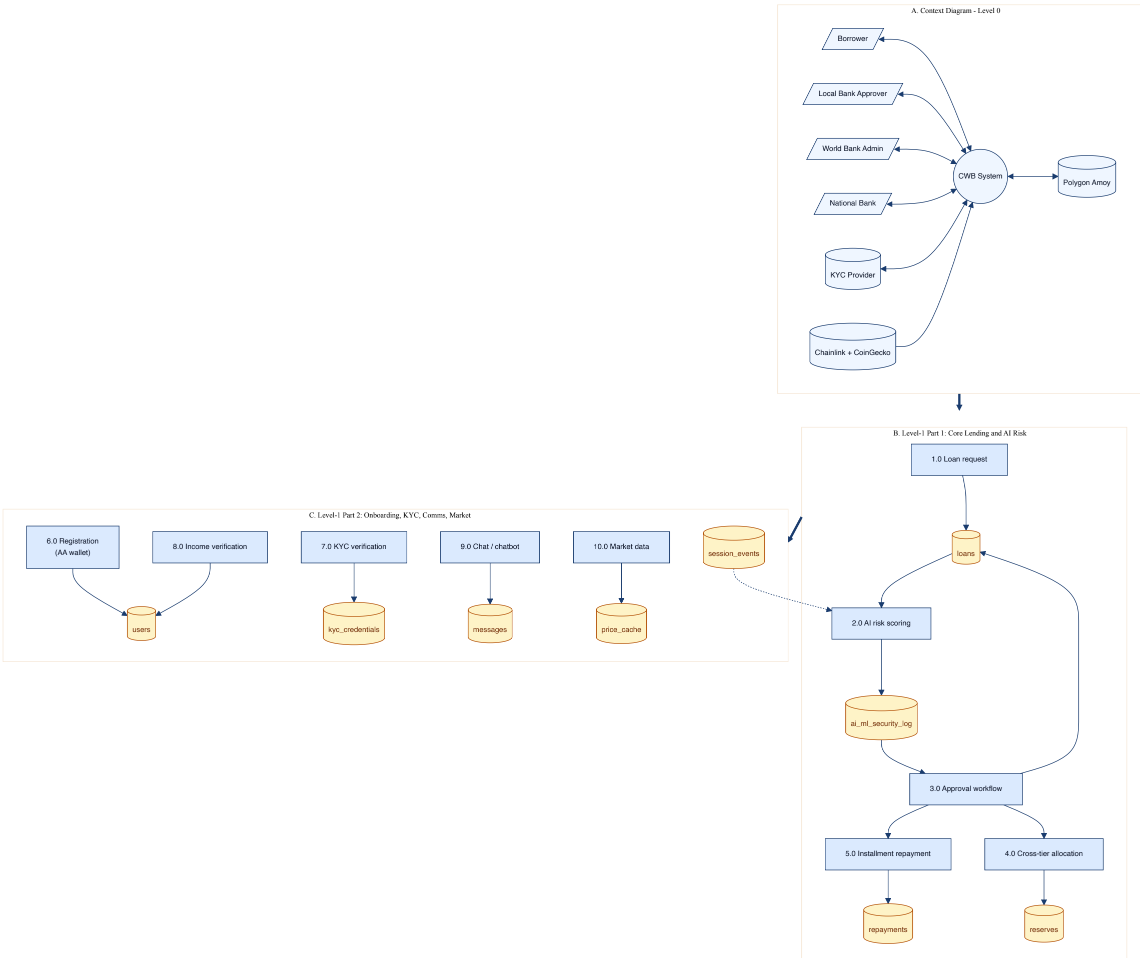



**Figure 3.19:** Consolidated DFD suite: Level-0 context plus two Level-1 partitions.

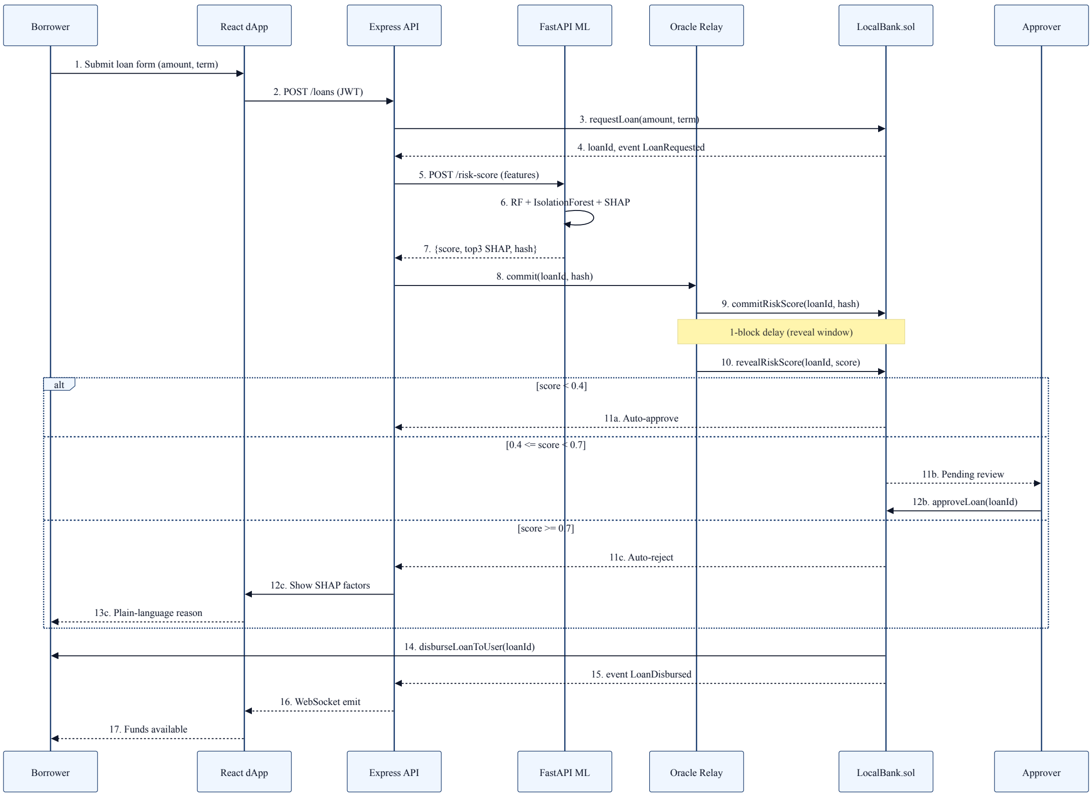



**Figure 3.20:** Loan approval sequence with three alt branches (auto-approve, manual review, auto-reject).

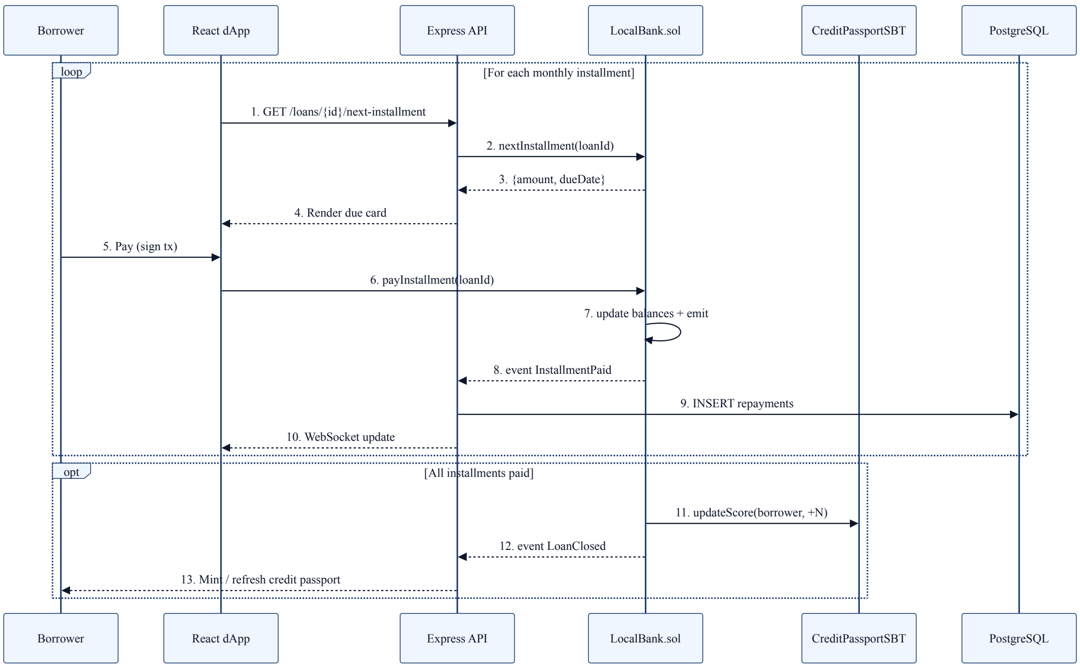



**Figure 3.21:** Installment payment loop with on-chain settlement, off-chain ledger update, and SBT update on full repayment.

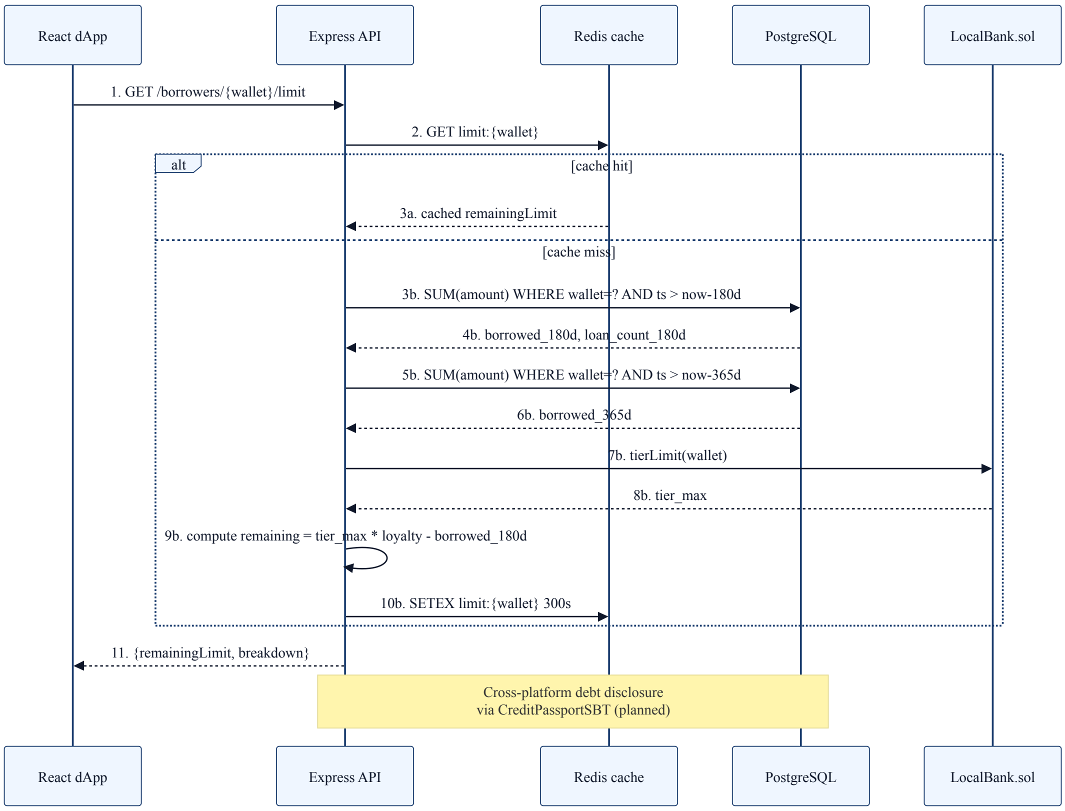



**Figure 3.22:** Borrowing-limit engine implementing formula F.19 with Redis cache and PostgreSQL window queries.

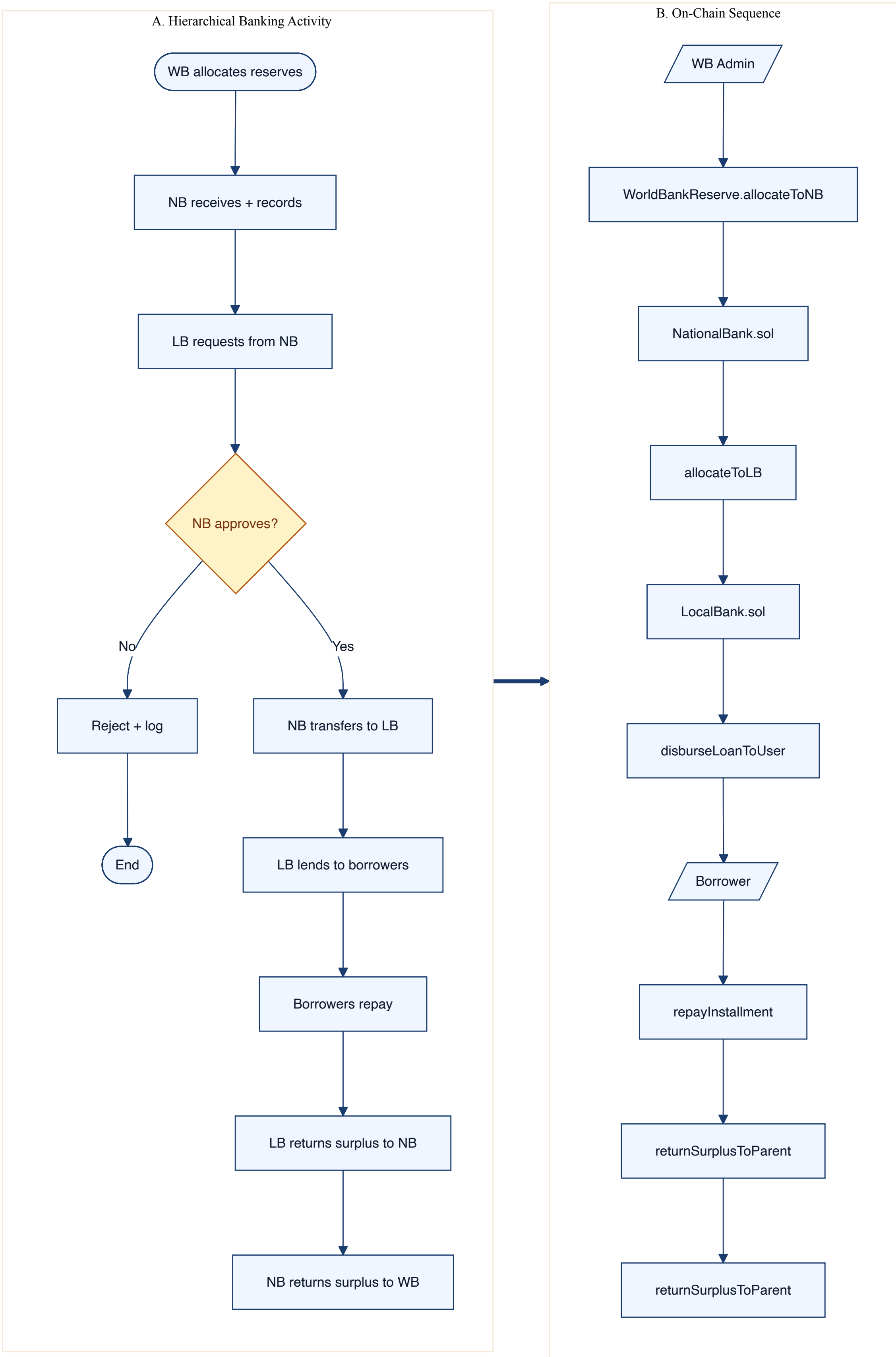



**Figure 3.23:** Hierarchical banking end-to-end: business activity (Panel A) and on-chain contract sequence (Panel B).

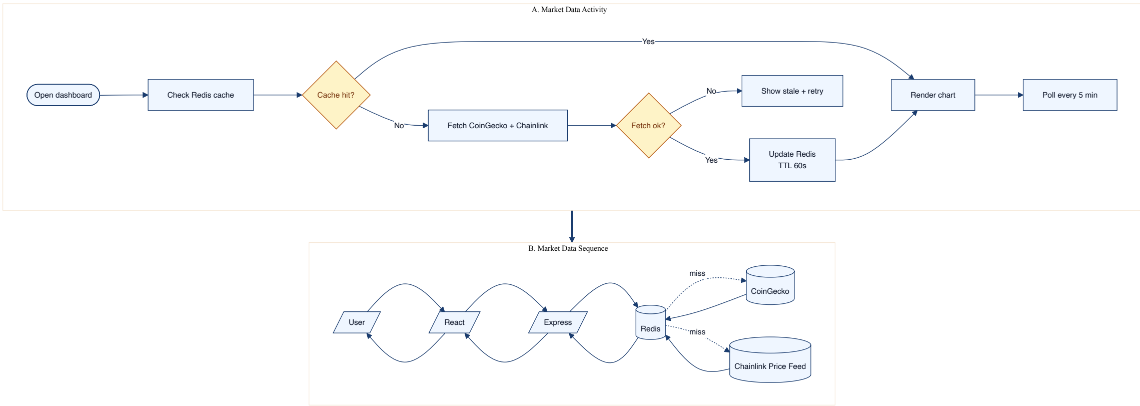



**Figure 3.24:** Market data: activity and sequence with Redis cache, CoinGecko, Chainlink Price Feed.

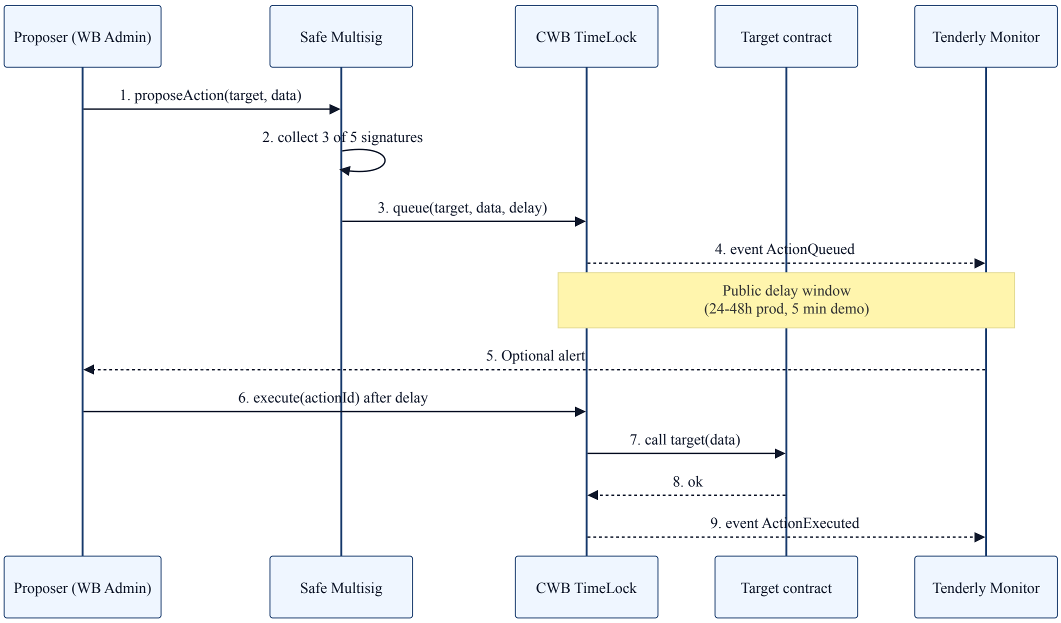



**Figure 3.25:** TimeLock governance sequence: propose, Safe multisig sign, queue with delay, execute.

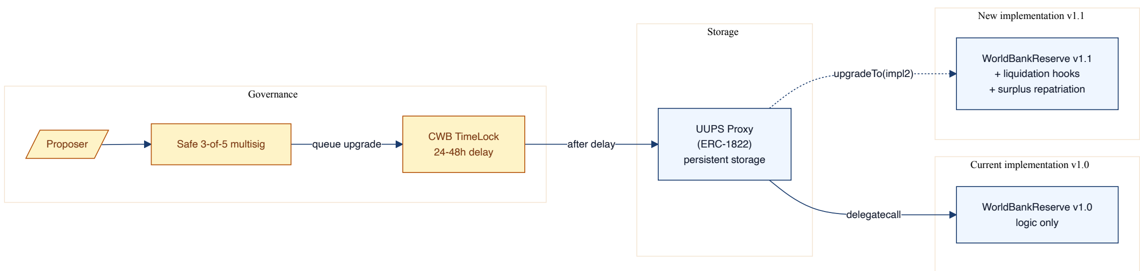



**Figure 3.26:** UUPS proxy upgrade path via Safe and TimeLock.

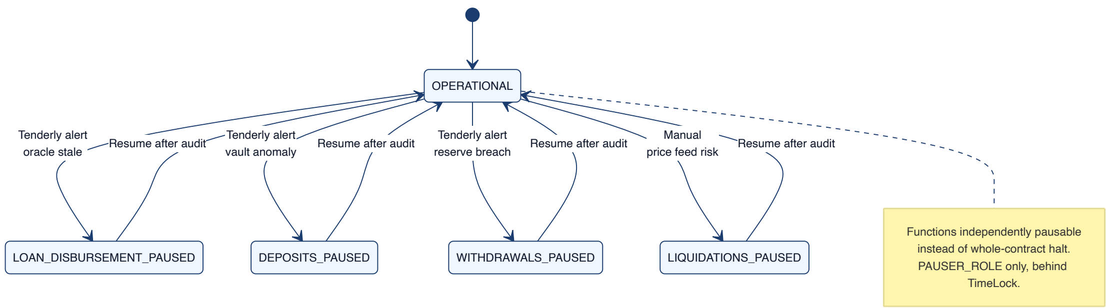



**Figure 3.27:** Granular pause states: individual functions pause without halting the contract.

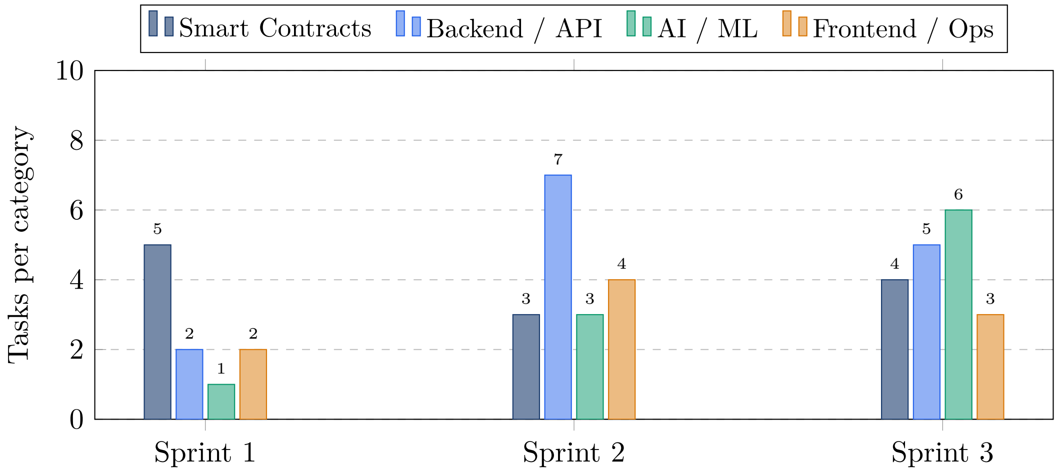



**Figure 4.1:** Sprint distribution by category. Sprint 1 is contracts-heavy (foundation), Sprint 2 expands the backend and frontend, Sprint 3 is AI / ML heavy and integration-heavy.

# Chapter 4

# Methodology

## 4.1 Development Methodology

The project follows an Agile/Scrum process with three sprints across pre-thesis 1 and pre-thesis 2. Pre-thesis 1 covers Sprints 1–2 (architecture, specification, partial scaffolding); pre-thesis 2 covers Sprint 3 (integration of AI/ML, liquidation engine, SavingsVault, zkAML circuit, formal verification pass). The Sprint distribution is shown in Figure 4.1, which consolidates the four separate pie charts of v13 into a single grouped bar chart (per-sprint breakdown), and the per-sprint backlog detail follows in Tables 4.2, 4.3, and 4.4.

## 4.2 Intelligent Services: Unified 9B Assistant

The platform provides three intelligent services through a single 9-billion-parameter base model fine-tuned with QLoRA: a borrower-facing chatbot, an internal security advisory channel surfaced on pull requests, and a risk-explanation service that translates SHAP feature contributions into plain language. Figure 4.2 shows the training pipeline (data ingestion, JSONL builder, QLoRA training, merge, GGUF export). Figure 4.3 consolidates the runtime view with three panels: the assistant's inference flow with ChromaDB RAG, the production scoring pipeline with RF + IF + SHAP, and the CI

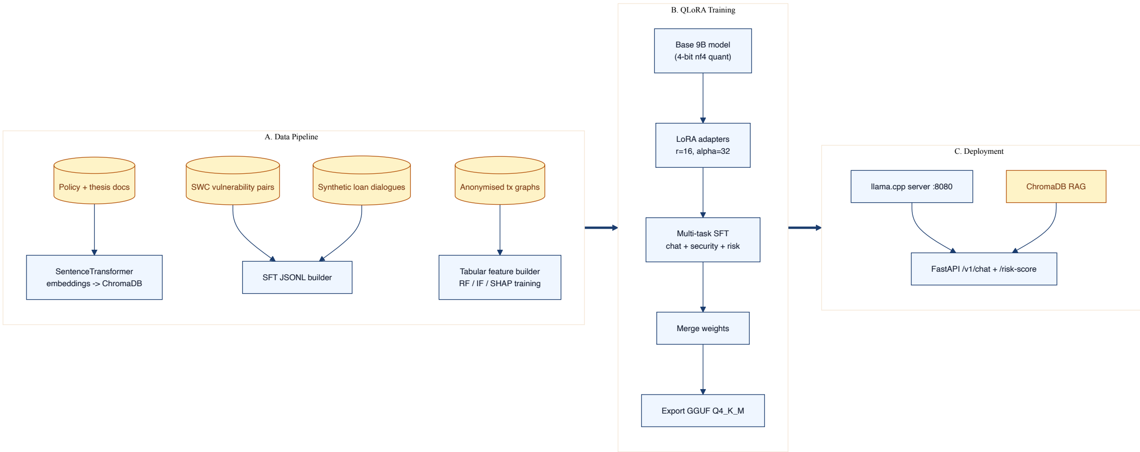



**Figure 4.2:** AI training pipeline: data ingest, JSONL builder, QLoRA SFT, merge, GGUF export.

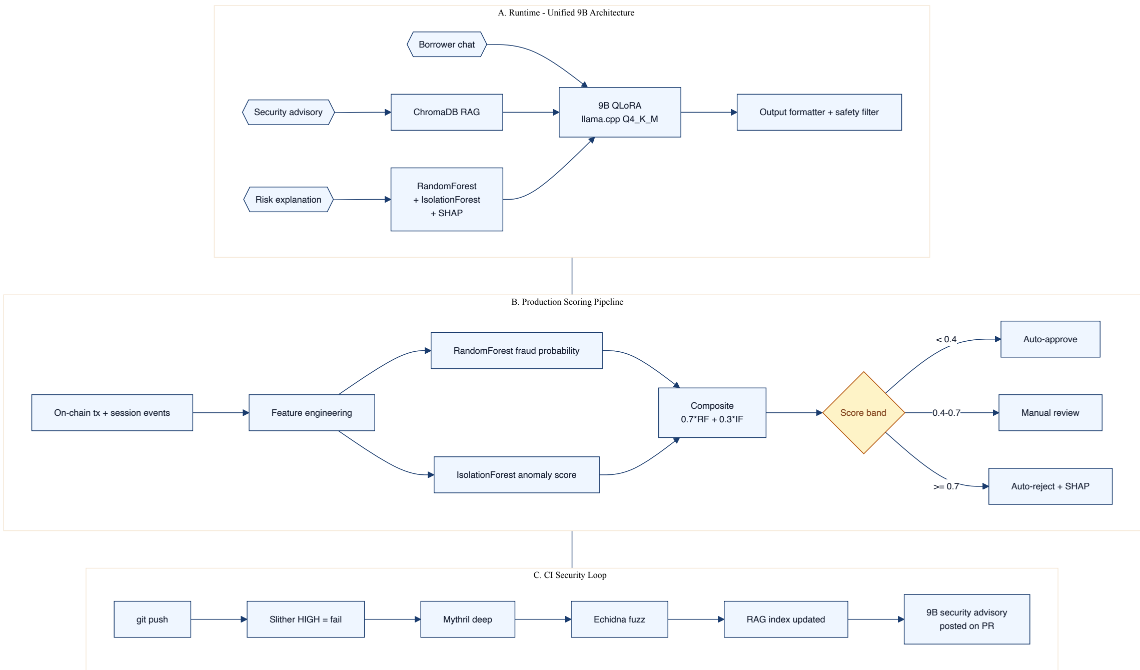



**Figure 4.3:** Runtime view: unified 9B assistant, production scoring pipeline, CI security loop.

security loop with Slither + Mythril + Echidna.

### 4.2.1 Base models and quantisation

The base model is a 7–9B class general LLM (Llama-3-8B-Instruct or Mistral-7B-v0.3) quantised to 4-bit NF4. The reasoning is empirical: 7–9B class models are the smallest that can sustain coherent multi-turn dialogue while still fitting on a single consumer GPU after 4-bit quantisation (memory $\approx 6$ GB). Export uses llama.cpp's GGUF

Q4_K_M format, which preserves ∼95% of full-precision accuracy on dialogue tasks at ∼4x throughput.

### 4.2.2 Task specification and guardrails

The assistant is trained on three task heads bundled into one model: (a) borrower dialogue (FAQs, repayment status, limit checks), (b) SWC vulnerability advisory (paired with vulnerable / patched Solidity snippets), and (c) SHAP narrative generation (feature contributions to plain English). Guardrails are: no investment advice; no automatic financial-decision execution; PII redaction in logs; constrained decoding for value-bound outputs (numeric limits, percentages).

### 4.2.3 Training data and QLoRA pipeline

Training data comes from four sources: (i) the policy and thesis corpus (this document, the BlockchainOlympiad guideline, World Bank publications); (ii) SWC vulnerability–patch pairs scraped from public audit reports; (iii) synthetic loan dialogues generated by GPT-4-class teacher models and then human-filtered; (iv) anonymised transaction graphs from Polygon testnet. The QLoRA configuration is LoRA $r = 16$, $\alpha = 32$, learning rate $2 \times 10^{-4}$, batch size 8, 3 epochs, warmup 5%.

### 4.2.4 Evaluation

The LLM is evaluated on three axes:

1. **Task accuracy:** F1 on a held-out 1k-example test set across the three task heads (target $\geq 0.85$).

2. **Safety:** a held-out red-team set covering injection, jailbreak, hallucination, and financial-advice prompts. The adversarial-vulnerability paper [llm(2025)] provides the threat catalogue. Target $\geq 0.95$ refusal accuracy on the red-team set.

3. **Calibration:** the SHAP narrative head is evaluated against a held-out set of borrower-graded explanations (clarity, accuracy) with the protocol of Bracke et al. [Bracke et al.(2019)].

## 4.3 Federated Learning for Multi-Bank Fraud Intelligence

The federated learning specification follows the PrivChain-AI [PrivChain-AI Authors(2025)] and Abbassi et al. [Abbassi et al.(2025)] designs: each Local Bank trains a local Random Forest + Isolation Forest model on its own data, and only the parameter deltas (not raw data) are sent to its parent National Bank, which aggregates via FedAvg (formula F.20). The aggregated regional model is then distributed back to all Local Banks. Cross-region aggregation at Tier 1 produces a global baseline. Figure 4.4 shows the topology.

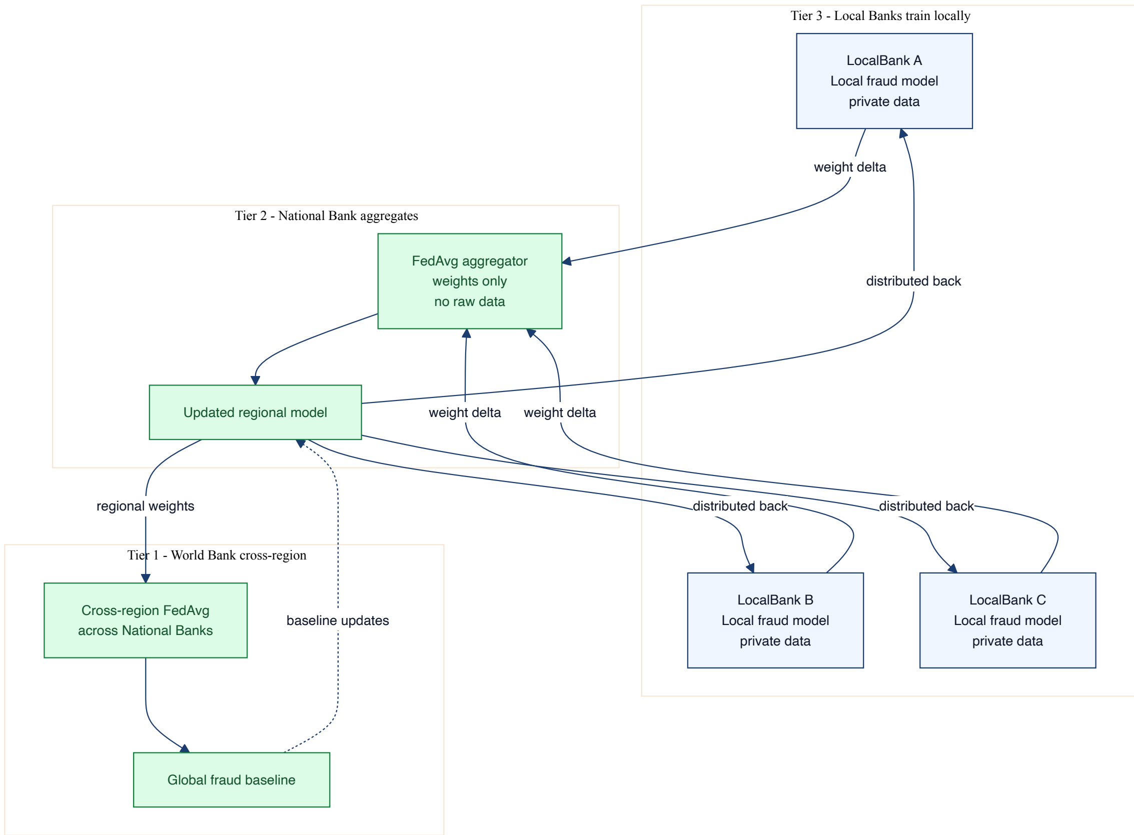



**Figure 4.4:** Federated learning topology: local training at Local Banks, FedAvg aggregation at National Banks, cross-region aggregation at the World Bank.

## 4.4 Graph Neural Network Extension

The Random Forest channel scores wallets independently. The 2024 GNN-on-DeFi-fraud literature [Cheng et al.(2024), Wang et al.(2024), Wang and Wang(2025)] demonstrates that wallet-graph neighbourhood signals (e.g., k-hop transaction structure) materially improve fraud detection. The pre-thesis 2 deliverable adds a graph layer: the wallet graph for each Local Bank is built from on-chain events indexed by The Graph subgraph; a GraphSAGE model computes per-wallet embeddings; the embedding becomes a feature consumed by the existing RandomForest. Ablation against the current tabular-only RF will be the empirical validation.

## 4.5 Formal Verification with Certora

Following the Certora 2025 open-source release [Certora(2025)], two CVL invariants are specified on the reserve contract:

1. **Reserve solvency.** `totalAllocated <= reserveBalance + sum(returnedSurplus)`: the WorldBankReserve never allocates more than it has received.

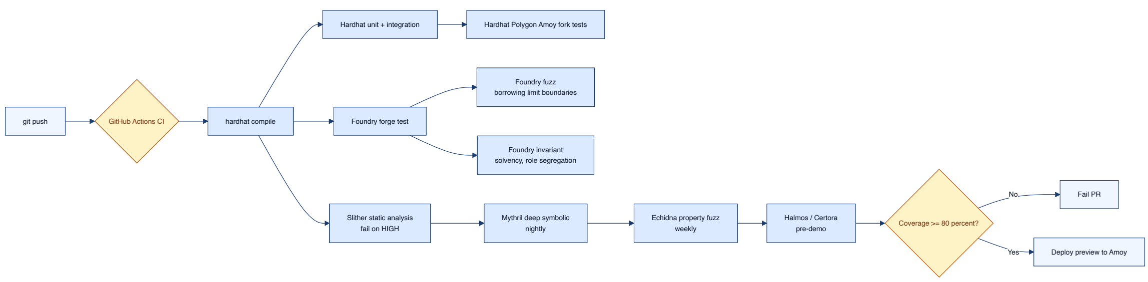



**Figure 4.5:** CI test pipeline: Hardhat unit + integration, Foundry fuzz + invariant, Slither static, Mythril deep, Echidna property, Halmos / Certora pre-demo.

2. **Role segregation.** No address can simultaneously hold both `LOAN_APPROVER_ROLE` and `BORROWER_ROLE` for the same loan.

These two properties cover the two highest-impact failure modes (insolvency at Tier 1 and approver self-dealing). Certora's CVL specification is sketched in Appendix C; the proof is part of pre-thesis 2.

## 4.6 Foundry Invariant and Fuzz Test Suite

The current scaffold has Hardhat unit tests only. Pre-thesis 2 introduces a Foundry invariant + fuzz suite covering:

- **Solvency invariant.** For every reachable state, sum of LocalBank balances $\leq$ sum allocated by NationalBanks.
- **Role-expiry invariant.** No expired role is honoured by any guarded function.
- **Rolling-limit fuzz.** Over a fuzzed sequence of loan requests, the cumulative 180-day disbursement to any borrower $\leq$ the tier ceiling at every moment.
- **HF boundary fuzz.** On randomly sampled $(C, D, LT, \text{price})$ quadruples, the liquidation engine triggers iff F.7 evaluates below 1.0.
- **Pause boundary fuzz.** Each granular pause flag suppresses its target function and only its target function.

The Foundry suite is integrated in the CI pipeline (Figure 4.5).

## 4.7 Agent-Based Economic Feasibility Simulation

A discrete-event agent-based simulation will validate the unit economics claimed in Chapter 5. Each agent class – World Bank, National Banks, Local Banks, retail borrowers, group borrowers, liquidators – has a behavioural rule. The simulation runs over

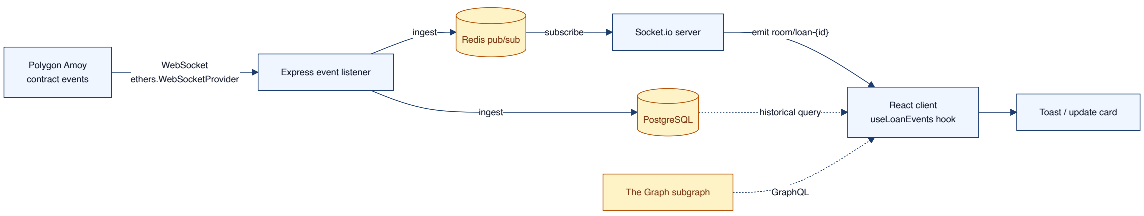



**Figure 4.6:** Real-time dashboard pipeline.

36 simulated months sweeping default rate (5%–15%) and average loan size (USD 100–1000); the output is per-tier annualised return, default-induced capital depletion, and liquidation throughput. The simulation is implemented in Mesa (Python) for pre-thesis 2.

## 4.8 Real-Time Dashboard Pipeline

Real-time visibility is critical for institutional confidence. The pipeline (Figure 4.6) ingests on-chain events via an ethers WebSocketProvider, persists them in PostgreSQL, publishes through a Redis pub/sub channel, and pushes to React clients over Socket.io. Historical queries fall back to PostgreSQL; portfolio summaries fall back to The Graph.

## 4.9 Transaction State Machine

Every borrower-initiated transaction (loan request, repayment, vault deposit) follows the state machine of Figure 4.7. Replaced transactions (gas-price-bumped by the wallet) are tracked through the new hash so the UI does not show false errors. Reverted transactions are surfaced with the decoded revert reason. This is part of the accessibility-design considerations of Section 5.14.

## 4.10 EIP-712 Authentication Sequence

Wallet-signed JWT replaces password authentication. Figure 4.8 shows the full challenge–sign–verify–JWT–blacklist flow. The JWT access token expires in 15 minutes; the refresh token expires in 7 days; logout blacklists the `jti` in Redis until natural expiry.

## 4.11 Evaluation Methodology

Evaluation is split into smart-contract validation, ML validation, and end-to-end integration validation. Table 4.1 fixes the quantitative targets.

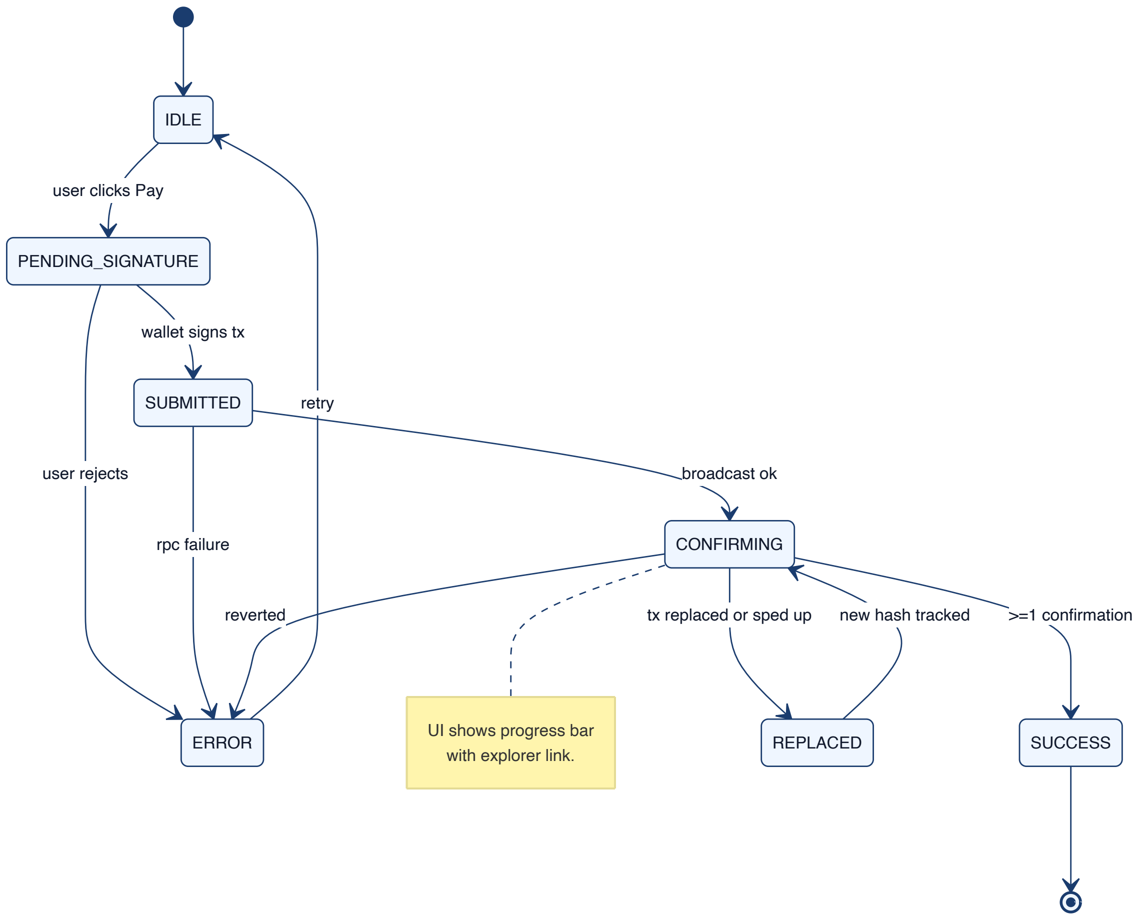



**Figure 4.7:** Transaction lifecycle state machine.

## 4.12 Sprint Plan and Deliverables

## 4.13 SDLC Stage Mapping

Pre-thesis 1 covers the requirements, design, partial implementation, and unit-test phases of the classical SDLC. Pre-thesis 2 covers integration, system testing, formal verification, and acceptance. The agile sprint structure is overlaid on this classical phase decomposition (Table 4.5).

## 4.14 Design Decisions and Alternatives

The principal design decisions and their alternatives are catalogued in Table 4.6. Each row pairs an alternative with the reason the chosen path was preferred.

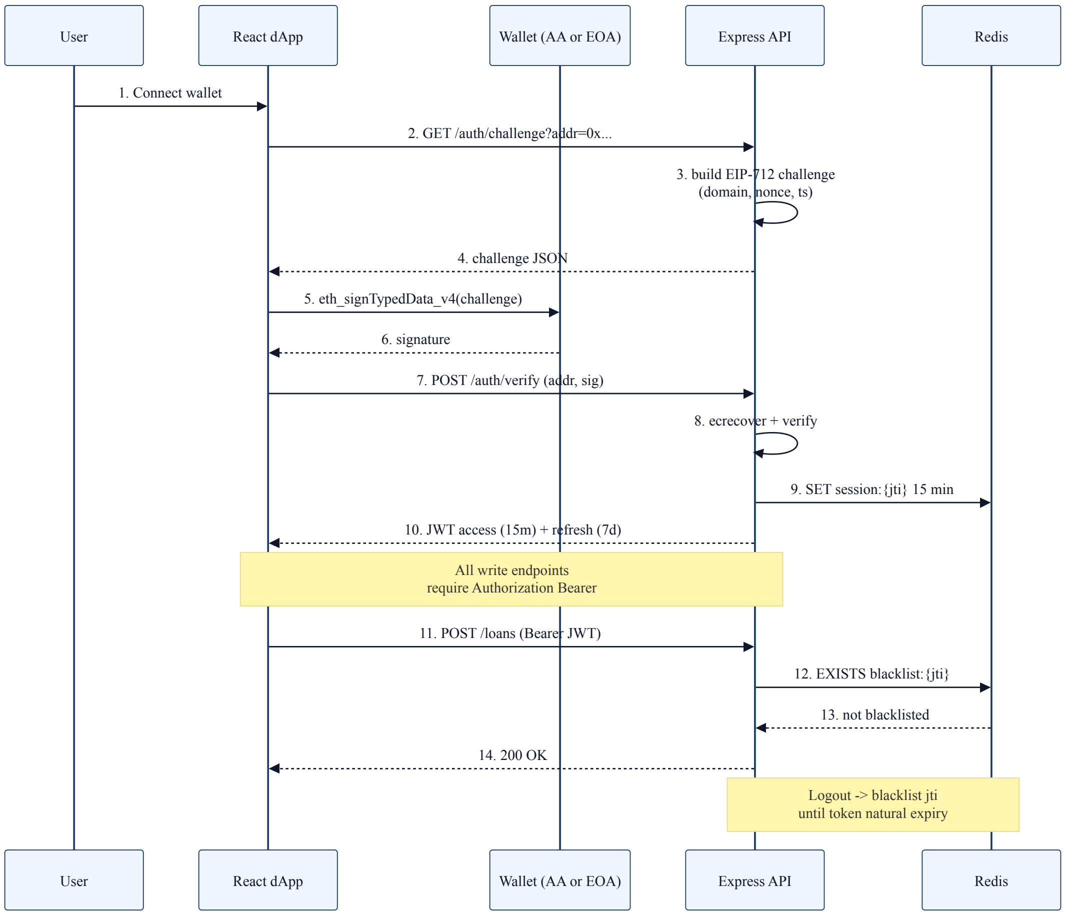



**Figure 4.8:** EIP-712 wallet-signed authentication: challenge, signature verification, JWT issuance, and Redis blacklist on logout.

## 4.15 Design Patterns

## 4.16 Software Testing Strategy

The testing strategy combines unit tests (Hardhat for contracts, Pytest for ML), property tests (Foundry invariants + fuzz), static analysis (Slither, Mythril, Echidna), formal verification (Certora CVL), integration tests on Polygon Amoy fork, and an end-to-end demo. The pipeline structure is Figure 4.5; coverage targets are in Table 4.1.

**Table 4.1:** Quantitative evaluation targets for the final thesis.

| Metric | Component | Target |
|---|---|---|
| Line + branch coverage | Smart contracts (Hardhat + Foundry) | $\geq$ 80% |
| Slither HIGH findings | Smart contracts | 0 |
| Mythril critical findings | Smart contracts | 0 |
| Echidna invariant violations | Smart contracts | 0 |
| Certora CVL proofs (reserve) | Smart contracts | 2 proven |
| ROC-AUC on RF fraud classifier | ML | $\geq$ 0.92 |
| F1 on RF fraud classifier | ML | $\geq$ 0.85 |
| Anomaly detection PR-AUC (IF) | ML | $\geq$ 0.85 |
| 9B assistant task F1 (3 heads avg) | LLM | $\geq$ 0.85 |
| 9B red-team refusal accuracy | LLM safety | $\geq$ 0.95 |
| End-to-end loan latency (request $\rightarrow$ disbursed) | E2E | $\leq$ 15 s |
| WebSocket round-trip latency (dashboard) | E2E | $\leq$ 250 ms |

**Table 4.2:** Sprint 1 backlog (architecture and contracts foundation).

| ID | Deliverable | Status |
|---|---|---|
| S1.1 | WorldBankReserve + allocation functions | ✓ |
| S1.2 | NationalBank scaffold + registration | ▷ |
| S1.3 | LocalBank scaffold + disbursement | ▷ |
| S1.4 | Loan + Repayment data model + SQL migrations | ✓ |
| S1.5 | PostgreSQL + Redis docker-compose | ✓ |
| S1.6 | EIP-712 challenge endpoint + JWT issue/verify | ✓ |
| S1.7 | React dApp shell + market data + AI chatbot stub | ▷ |
| S1.8 | Improved mermaid diagram set (43 figures) | ✓ |
| S1.9 | Pre-thesis v14 LaTeX overhaul | ✓ |

**Table 4.3:** Sprint 2 backlog (AI/ML and governance specification).

| ID | Deliverable | Status |
|---|---|---|
| S2.1 | FastAPI ML service: RF + IF + SHAP | ▷ |
| S2.2 | 9B QLoRA training data preparation | ○ |
| S2.3 | LiquidationEngine specification (this chapter) | ✓ |
| S2.4 | SavingsVault + FixedDeposit specification | ✓ |
| S2.5 | Borrower limit engine + Redis cache | ▷ |
| S2.6 | The Graph subgraph schema | ○ |
| S2.7 | Safe multisig + TimeLock deployment script | ○ |
| S2.8 | MiCA + GENIUS Act compliance mapping | ✓ |
| S2.9 | Foundry test scaffold + 5 invariants | ○ |
| S2.10 | ChromaDB RAG ingestion of policy corpus | ○ |

**Table 4.4:** Sprint 3 backlog (integration + verification for pre-thesis 2).

| ID | Deliverable | Status |
|---|---|---|
| S3.1 | Commit–reveal oracle wiring (LoanController) | ○ |
| S3.2 | Chainlink Functions migration | ○ |
| S3.3 | KYCVerifier Circom 2.0 circuit + integration | ○ |
| S3.4 | AMLVerifier zkAML circuit + Merkle root refresh | ○ |
| S3.5 | CreditPassportSBT (ERC-5114-like) | ○ |
| S3.6 | Federated FedAvg aggregator (Tier 2) | ○ |
| S3.7 | GraphSAGE ablation against tabular RF | ○ |
| S3.8 | Certora CVL proofs (reserve solvency, role segregation) | ○ |
| S3.9 | Tenderly monitor + 6 alert triggers configured | ○ |
| S3.10 | Bridge candidate evaluation (LayerZero / Axelar / CCIP) | ○ |
| S3.11 | Mesa agent-based feasibility simulation | ○ |
| S3.12 | End-to-end demo on Polygon Amoy | ○ |

**Table 4.5:** SDLC stage mapping.

| Phase | Sprint | Output |
|---|---|---|
| Requirements | S1 | Stakeholder analysis, use cases, BB SP2025-02 grounding |
| Design | S1, S2 | Architecture (Ch. 3), 43 improved diagrams |
| Implementation (partial) | S1, S2 | Reserve, NB, LB scaffolds; FastAPI ML; React dApp |
| Unit testing | S2 | Hardhat unit tests, Pytest ML tests |
| Integration testing | S3 | Hardhat fork tests + Foundry invariants |
| Formal verification | S3 | Certora CVL proofs |
| System / acceptance testing | S3 | End-to-end demo, agent simulation |

**Table 4.6:** Design decisions and alternatives.

| Decision | Chosen vs. alternative | Rationale |
|---|---|---|
| L1 chain | Polygon Amoy vs. Ethereum mainnet | USD 0.01–0.05 gas per op vs. USD 5–15; PoS finality ∼5s; mainnet deferred |
| Loan denomination | Stablecoin vs. native ETH | MiCA + GENIUS; eliminates volatility doubling real debt in drawdown |
| Identity layer | DID + VC + ZKP vs. raw on-chain PII | SSI; GDPR / PDPA compliance; on-chain stores flag only |
| Upgrade pattern | UUPS vs. Transparent Proxy vs. Beacon | Cheaper proxy calls; upgrade logic in implementation; auditable via TimeLock |
| Multisig | Safe 3-of-5 vs. single EOA | Single-key compromise unacceptable; 3-of-5 supports staff rotation |
| Risk pipeline | Commit-reveal oracle vs. direct on-chain ML | On-chain ML infeasible; commit-reveal prevents relay manipulation |
| ML model | RF + IF + SHAP vs. pure neural | SHAP-friendly; interpretable; small training set sufficient |
| LLM | 9B QLoRA vs. API call to GPT-4 | Cost predictability; on-prem privacy; deterministic version |
| KYC provider | Veriff vs. Persona vs. Sumsub | Bangladesh coverage; sandbox-compatible API; price competitive |
| SBT standard | ERC-5114-like vs. ERC-721 transferable | Non-transferable is the point; resists capture and sale |
| Bridge | Defer decision to pre-thesis 2 | 2022–2024 bridge hacks history; need empirical evaluation |
| Liquidation incentive | 5–8% keeper bonus vs. DAO fund | Permissionless keeping; aligns with Aave V3 norms |

**Table 4.7:** Software design patterns applied.

| Pattern | Application in CWB | Layer |
|---|---|---|
| Proxy (UUPS) | WorldBankReserve, LB, NB upgradeable | L1 contract |
| Factory | LocalBankContract deployed per Local Bank | L1 contract |
| Observer | The Graph subgraph subscribes to events | L4 indexing |
| State Machine | Loan lifecycle (Figure 4.7) | L1 + L7 |
| Strategy | RF + IF composite scoring | L3 ML |
| Adapter | CWBPriceFeed wraps Chainlink + CoinGecko | L1 contract |
| Repository | DAOs over PostgreSQL in Express | L2 application |
| Singleton | WorldBankReserve (single global) | L1 contract |
| Circuit Breaker | Granular pause states (Figure 3.27) | L1 contract |
| Saga | Cross-tier allocation with compensating reverts | L1 + L2 |

# Chapter 5

# Market Analysis and Feasibility

## 5.1 Market Sizing

The funnel of Figure 1.1 resolves to: **TAM** (Total Addressable Market) of USD 18 trillion across global SME finance, remittances, and microcredit; **SAM** (Serviceable Addressable Market) of USD 850 billion across emerging-market mobile-first credit; **SOM** (Serviceable Obtainable Market) of USD 14 billion across Bangladesh + South Asia digital-lender share over five years. The Bangladesh-specific anchor is the 41.56 million microfinance accounts reported by Bangladesh Bank [Bangladesh Bank(2025b)] and the women-led BRAC microfinance reach reported in 2025 [BRAC(2025)].

## 5.2 Target Customer Segment

The primary customer segment is women-led solidarity-group microbusinesses in Bangladesh's rural and peri-urban districts (10–12 borrower groups around each Local Bank operator). The secondary segment is SMEs that fall through MFI ceilings but below commercial-bank floors – the USD 3,000–30,000 monthly turnover band – with documented MSME credit gap of USD 5.2 trillion globally [International Finance Corporation(2019)] and Bangladesh share approximately USD 150 billion. The tertiary segment is institutional allocators (development banks, donor agencies) channeling capital through Local Banks.

## 5.3 Partner Ecosystem

Five partner classes are required to reach the customer segments above: (i) *Local Bank operators* (BRAC Bank pilot, Eastern Bank pilot, ASA digital pilot); (ii) *KYC providers* (Veriff, Persona, Sumsub) – evaluated on Bangladesh NID coverage; (iii) *stablecoin issuers / on-ramps* (Circle USDC, Tether USDT, Stellar XLM USDC pair via Stellar Disbursement Platform [Stellar Development Foundation(2025)]); (iv) *infrastructure* (Polygon Foundation, Chainlink, Alchemy / Biconomy for AA, Tenderly, OpenZeppelin Defender); (v) *academic / NGO partners* (BRAC Microfinance, BRAC Institute of Governance and Development, Grameen Foundation).

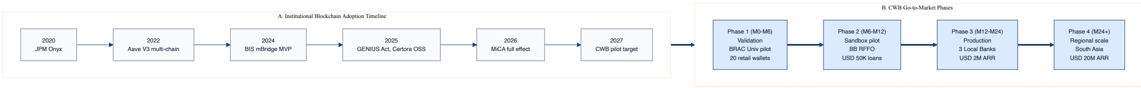



**Figure 5.1:** Institutional-blockchain adoption timeline and CWB phased go-to-market.

## 5.4 Phased Go-to-Market

Figure 5.1 situates CWB in the institutional-blockchain adoption timeline (JPM Onyx 2020, Aave V3 multi-chain 2022, BIS mBridge MVP 2024, MiCA full effect 2026, CWB pilot target 2027) and decomposes the deployment into four phases: validation (BRAC University pilot, 20 retail wallets), sandbox (Bangladesh Bank RFFO, USD 50K loans), production (three Local Banks, USD 2M ARR), and regional scale (South Asia, USD 20M ARR).

## 5.5 Competitive Landscape

Figure 5.2 shows the four-quadrant positioning of CWB against eleven competitors. The unique position is the top-right (institutional + multi-tier) quadrant.

Table 5.1 consolidates the three separate competitor tables of v13 (Direct DeFi, TradFi institutional, regional Asia/microfinance) into a single wide booktabs table with thirteen rows.

## 5.6 Risk Taxonomy

Table 5.2 consolidates technical, market, and regulatory risks with mitigation.

## 5.7 Technical Feasibility

Per-loan gas cost on Polygon Amoy is dominated by storage writes: loan creation $\approx$ 230k gas, repayment $\approx$ 95k gas, liquidation $\approx$ 320k gas. At Polygon's gas-token price ($\sim$0.05 GWEI MATIC, $\sim$USD 0.50/MATIC), one full loan lifecycle costs less than USD 0.05 – two orders of magnitude below the spread captured by the platform. ML inference per scoring call is $\sim$120 ms on a single CPU core with the FastAPI service; the QLoRA assistant returns first token within 1.2 s on consumer-grade GPU.

## 5.8 Economic Feasibility

Net interest allocation (formula F.16) under the kinked rate model (Section 3.8) for a representative Local Bank with USD 100k portfolio at 65% utilisation yields a $\sim$2% net spread after depositor yield and 0.5% insurance allocation. Per the agent-based simulation (Section 4.7) at 8% default rate, capital recovery from the LiquidationEngine

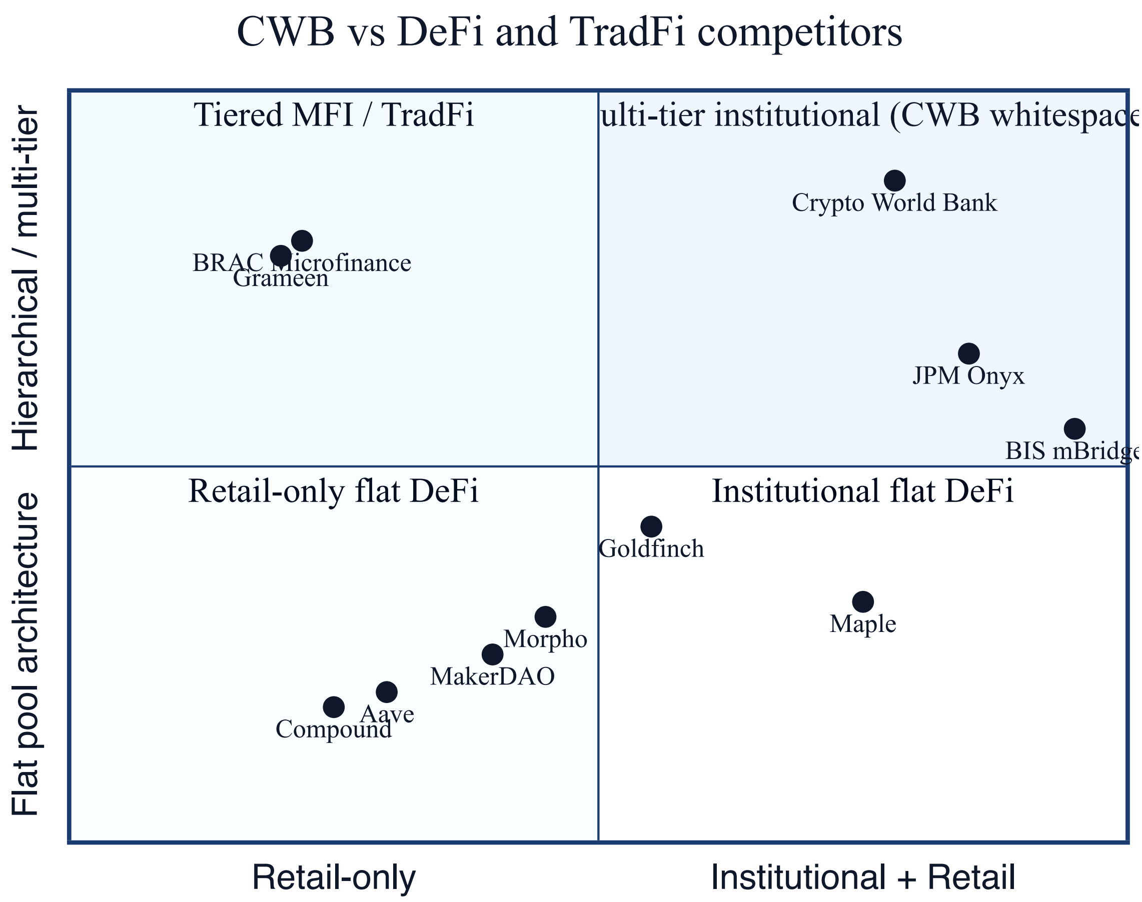



**Figure 5.2:** Quadrant chart of competitors. The CWB position is institutional + multi-tier (top right).

+ insurance pool keeps Local Bank capital depletion below 1.5% per annum – below the spread.

## 5.9 Revenue Projection

Figure 5.3 (refurbished from v13) shows the projected annual revenue by tier under steady-state operating assumptions of the agent-based simulation; Figure 5.4 (refurbished from v13) shows the hierarchical interest-rate spread; Figure 5.5 (new in v14) shows the revenue sensitivity to default rate and average loan size.

## 5.10 Currency Risk and Stablecoin Imperative

The volatile-collateral analysis of v13 is extended in v14 with the explicit MiCA + GENIUS compliance mapping (Section 3.3.1 and Appendix E). The empirical justification for the stablecoin-first design is the May 2021 and 2022 ETH drawdowns of 50% within months, doubling real borrower debt where loans were denominated in ETH.

**Table 5.1:** Consolidated competitive landscape (DeFi, TradFi institutional, regional / microfinance).

| Competitor | Class | TVL or AUM (2026) | Strength | Gap |
|---|---|---|---|---|
| Aave V3 [DefiLlama(2026a)] | DeFi | USD 25B | Composability, mature | Flat pool; no hierarchy |
| Compound V3 [DefiLlama(2026b)] | DeFi | USD 4B | Audited, stable | Flat pool; no hierarchy |
| Morpho [Morpho(2026)] | DeFi | USD 12B | Vault customisation | No institutional governance |
| Sky / MakerDAO [Fensory(2026c)] | DeFi | USD 20.6B | Stablecoin issuance | Single-tier |
| Maple Finance [Fensory(2026b)] | DeFi (institutional) | USD 2B | Institutional pools | No retail tier |
| Goldfinch | DeFi (RWA) | USD 200M | RWA collateral | No hierarchical reserve |
| Centrifuge [Centrifuge(2025)] | DeFi (RWA) | USD 700M | RWA tokenisation | No retail tier |
| JPM Onyx (Kinexys) [JPMorgan Chase(2025)] | TradFi | USD 1.5T settled | Institutional grade | Closed network |
| BIS mBridge [Bank for International Settlements(2024c)] | TradFi | MVP | Sovereign CBDC bridge | Settlement only; no lending |
| Project Agora [Bank for International Settlements(2024b)] | TradFi | Pilot | Unified ledger | Settlement only |
| Grameen Bank | MFI | USD 1.2B | Solidarity model proven | Off-chain; no portability |
| BRAC Microfinance [BRAC(2025)] | MFI | USD 4B | Reach + women focus | Off-chain; no portability |
| **Crypto World Bank** | DeFi + MFI hybrid | — | Multi-tier + AI + ZKP + SBT | Pilot scale (this work) |

## 5.11 Bootstrap Funding and Tier 1 Capitalization

## 5.12 Over-Indebtedness Risk in Group Lending

The NUS 2025 microfinance analysis [NUS Economics Society(2025)] documents that with 724 licensed MFIs in Bangladesh serving 41.56 million accounts, individual borrowers frequently service simultaneous loans from multiple institutions – a structural over-indebtedness risk that solidarity-group guarantees alone do not solve. CWB's two mitigation controls are (a) the CreditPassportSBT exposing cumulative active debt across compatible lenders (Section 3.6.4), allowing any querying lender to refuse a loan that would push total debt beyond a sustainable threshold; and (b) a hard on-chain cap on the number of simultaneously active loans per borrower (default 2; configurable

**Table 5.2:** Risk taxonomy with mitigation.

| Risk class | Specific risk | Mitigation |
|---|---|---|
| Technical | Smart contract bug | 4-tool audit pipeline + Certora CVL proofs |
| Technical | Oracle manipulation | Commit-reveal + Chainlink Price Feed + staleness check |
| Technical | Bridge hack | Single-chain pre-MVP; bridge evaluation in pre-thesis 2 |
| Market | Stablecoin de-peg [sta(2025)] | Diversify across USDC + USDT + planned local-CBDC |
| Market | Borrower default cluster | Solidarity-group joint liability + 5–8% liquidation bonus |
| Market | Over-indebtedness (cross-platform) [NUS Economics Society(2025)] | CreditPassportSBT + cap on simultaneous active loans |
| Regulatory | MiCA non-compliance | 1:1 reserve + authorisation + audit (Section 3.3.1) |
| Regulatory | GENIUS Act non-compliance | USDC integration + licensed issuer custody |
| Regulatory | Bangladesh sandbox rejection | Pilot scope aligned with FE Circular No. 06 + RFFO sandbox |
| Operational | Safe signer compromise | 3-of-5 multisig + hardware + key rotation policy |
| Operational | ML data drift | Quarterly re-training + Tenderly drift alerts |

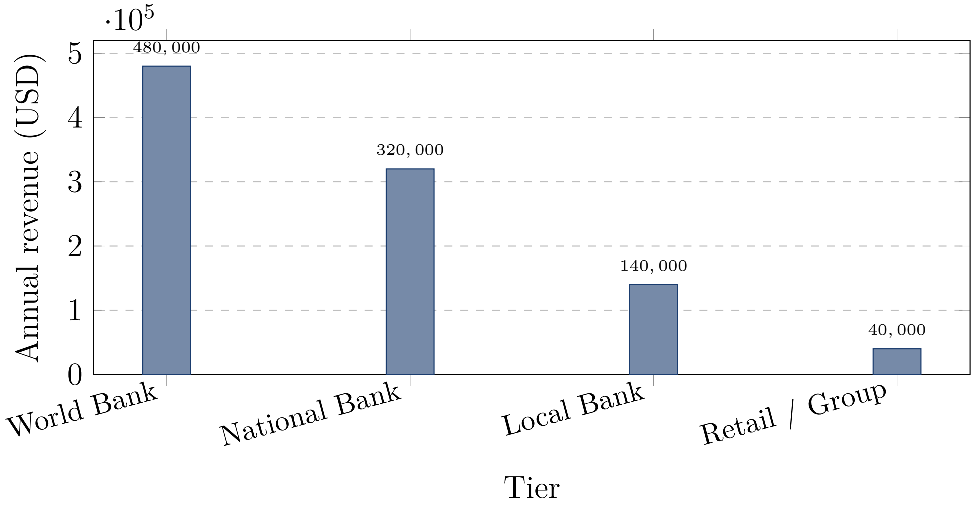



**Figure 5.3:** Projected annual revenue by tier under the agent-based simulation steady state. Year-3 production scale, three Local Banks, USD 2M aggregate ARR.

per Local Bank). Both controls are unenforceable in the current paper-based MFI ecosystem and are the principal social-impact contribution of the architecture.

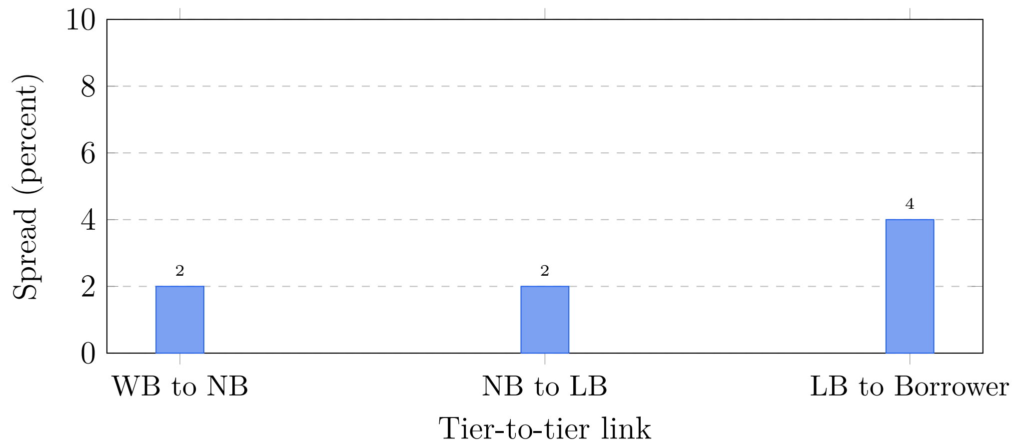



**Figure 5.4:** Hierarchical interest-rate spread: cumulative borrower APR of ~8% is consistent with the waterfall in Figure 1.6.

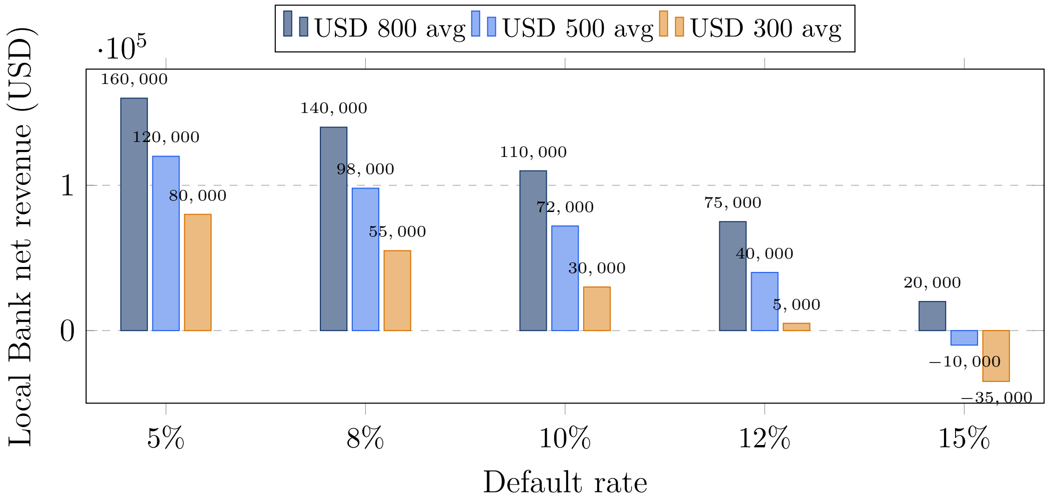



**Figure 5.5:** Revenue sensitivity to default rate and average loan size for a representative Local Bank. The USD 300 average loan turns unprofitable above 12% default; the over-indebtedness controls of Section 5.12 keep portfolio default below this band.

## 5.13 Prototype Scope and Limitations

The pre-thesis 1 deliverable is bounded by: (a) Polygon Amoy testnet only, no mainnet; (b) test stablecoins, no production stablecoin custody; (c) mock KYC documents, no production KYC provider integration; (d) the BRAC University pilot cohort only, no public users; (e) simulated AI risk scores, full ML pipeline integration in pre-thesis 2.

## 5.14 Accessibility Design Considerations

Every figure in v14 carries a `\Description{}` alt-text body (the requirement on the macro is enforced in Section 3.2). Tables are designed with sufficient horizontal whitespace and no decorative row shading per ACM convention. The React dApp inherits Tailwind's accessibility tokens (focus rings, semantic colour, ARIA roles). The trans-

**Table 5.3:** Funding sources for Tier 1 capitalization.

| Source | Stage | Notes |
|---|---|---|
| University seed grant | Pilot | BRAC University CSE seed pool (in-kind) |
| Polygon Foundation grant | Pilot | Polygon Village programme for civic apps |
| DAO treasury (Aave Grants DAO) | Sandbox | Aave grants for Aave-V3-compatible protocols |
| Stellar Community Fund [Stellar Development Foundation(2025)] | Sandbox | Microfinance focus + Stellar Disbursement |
| World Bank Innovation Hub | Production | Project-level grant through MDB programmes |
| Donor cohort (BMGF, USAID) | Production | Microfinance digitisation focus area |
| National Bank pilot equity | Production | Local Bank operator co-investment |

action state machine (Figure 4.7) is a deliberate accessibility design choice: a clear UI for "pending", "confirming", "success", "replaced", "error" states reduces cognitive load for first-time crypto users.

## 5.15 Statement of Contributions

**Pre-thesis 1.** M.B.R.J. and M.M.A.A. jointly developed the four-tier architecture, the diagram and table consolidation strategy, the bibliography migration, and the literature review. M.B.R.J. led the smart-contract specification (reserve, banks, liquidation engine, SavingsVault, TimeLock, SBT) and the on-chain / off-chain partitioning matrix. M.M.A.A. led the AI/ML pipeline specification (RF + IF + SHAP + 9B QLoRA + federated learning), the EIP-712 authentication flow, and the methodology chapter. Both authors jointly reviewed the regulatory compliance mapping. The supervisor, Mr. Annajiat Alim Rasel, provided iterative feedback on diagram consistency and the institutional-DeFi positioning. No external consultants were employed.

# Chapter 6

# Conclusion

Crypto World Bank is a complete institutional banking architecture specified for implementation on programmable blockchain infrastructure. The four-tier hierarchical lending model, the stablecoin-first loan denomination, the AI risk pipeline wired through a commit–reveal oracle, the dual ZKP compliance gateway covering both identity and sanctions, the soulbound credit passport, the LiquidationEngine, the SavingsVault, and the federated learning specification together address the gaps that the institutional-DeFi report [Sygnum Bank(2026)] identifies and that the Bangladesh microfinance evidence base 2025 [Bangladesh Bank(2025b), BRAC(2025), NUS Economics Society(2025)] grounds in current local data. Five novel research contributions distinguish this work: the dual ZKP-KYC + ZKP-AML circuit, the SBT-based portable credit history, the federated multi-bank fraud intelligence, the explicit MiCA + GENIUS compliance mapping with a Bangladesh sandbox pathway, and the formal-verification Certora CVL specification on the reserve solvency invariant.

## 6.1 Future Work

Twelve concrete deliverables remain for pre-thesis 2 and the final thesis:

1. Integrate the LoanController commit–reveal oracle end-to-end with FastAPI ML.
2. Migrate the oracle relay to Chainlink Functions.
3. Implement the LiquidationEngine and exercise it against an adversarial fuzz suite.
4. Implement the SavingsVault + FixedDeposit with reserve-ratio gated withdrawal.
5. Implement the KYCVerifier Circom 2.0 circuit and the AMLVerifier zkAML circuit.
6. Implement the CreditPassportSBT and the cross-platform attestation interface.
7. Implement the federated FedAvg aggregator across three pilot Local Banks.
8. Train and evaluate the 9B QLoRA assistant against the targets of Table 4.1.
9. Add the GraphSAGE feature channel and ablate against the tabular RF.

10. Prove the two Certora CVL invariants on the reserve contract and publish the proofs.
11. Run the Mesa agent-based feasibility simulation across the default-rate / loan-size grid.
12. Apply for Bangladesh Bank RFFO sandbox admission with a Local Bank partner.

# Appendix A

# Technology Stack Detail

**Table A.1:** Refreshed technology stack with all new v14 modules.

| **Layer** | **Component** | **Selection rationale** |
|---|---|---|
| L1 Network | Polygon Amoy | PoS, 5s finality, low gas, EVM-compatible |
| L2 Contracts | Solidity 0.8.x | Native EVM; checked arithmetic |
| | UUPS proxies (ERC-1822) | Cheaper than transparent; auditable upgrade |
| | Safe multisig | Industry standard institutional custody |
| | OpenZeppelin (AccessControl, Pausable, ReentrancyGuard) | Audited primitive library |
| | CWB TimeLock | 24–48h delay on governance actions |
| L3 Storage | PostgreSQL 16 | ACID, JSONB for SHAP, mature |
| | Redis 7 | Cache, JWT blacklist, pub/sub |
| | S3 / MinIO | Encrypted KYC docs, off-chain |
| L4 Index/Monitor | The Graph | Industry-standard event indexer |
| | Tenderly | Production-grade monitor + alerts |
| | OpenZeppelin Defender | Automated upgrade + monitor |
| L5 AI/ML | FastAPI + scikit-learn (RF, IF) | Lightweight, well-supported |
| | SHAP | Reference XAI library [Lundberg and Lee(2017)] |
| | llama.cpp + GGUF Q4_K_M | On-prem LLM serving |
| | ChromaDB | RAG vector index |
| L6 Orchestration | Express.js + Socket.io | WebSocket real-time + REST |
| L7 Client | React + Tailwind | Mature, accessible |
| | ethers.js + wagmi | EVM client SDK |
| | Biconomy / Alchemy AA | ERC-4337 SmartAccount |
| External | Chainlink Price Feeds + Functions | Decentralised oracle |
| | Veriff / Persona | KYC provider with BD coverage |
| | Firebase Auth | Email + social auth |

**Table B.1:** Smart contract capability matrix with new v14 modules.

| Contract | Purpose | Status |
|---|---|---|
| WorldBankReserve | Tier 1 reserve, allocation, surplus reception, reserve invariants | ✓ |
| NationalBankContract | Tier 2 institutional borrower / allocator | ▷ |
| LocalBankContract | Tier 3 retail lending, disburseLoanToUser, repayments | ▷ |
| LoanController | Commit–reveal oracle + risk-score storage | ○ |
| LiquidationEngine | Health Factor enforcement + keeper bonus | ○ |
| SavingsVault | Variable-APY pool, reserve-ratio gated withdraw | ○ |
| FixedDeposit | Duration-matched liability | ○ |
| KYCVerifier | Circom 2.0 ZKP-KYC circuit verifier | ○ |
| AMLVerifier | zkAML circuit verifier + Merkle root setter | ○ |
| CreditPassportSBT | Non-transferable credit history token | ○ |
| DIDRegistry | W3C DID anchor + key rotation | ○ |
| CWBTimeLock | TimeLock for all admin actions (24–48h) | ○ |
| CWBPriceFeed | Chainlink adapter + staleness check | ○ |
| SolidarityGroup | Group lending contract (per-group factory deploy) | ○ |
| FXModule | Stablecoin / FX conversion with slippage | ○ |
| TradeFinanceLC | Letter-of-credit (final thesis) | × |

# Appendix B

# Smart Contract Capability Matrix

# Appendix C

# Certora CVL Specification Sketch

The two Certora CVL invariants for the reserve contract, sketched in CVL syntax for the pre-thesis 2 verification pass:

```
invariant reserveSolvency()
    totalAllocated() <= reserveBalance() + sumReturnedSurplus();

rule allocationPreservesSolvency() {
    env e; uint256 amount; address target;
    require allocateToNationalBank(e, amount, target);
    assert reserveSolvency();
}
```

**Listing C.1:** Reserve solvency invariant.

```
invariant noApproverIsBorrower(address u, uint256 loanId)
    !(hasRole(LOAN_APPROVER_ROLE, u) && getBorrower(loanId) == u);
```

**Listing C.2:** Role segregation invariant.

# Appendix D

# WorldBankReserve Public Interface (Partial Listing)

```
function allocateToNationalBank(uint256 amount, address nb) external;
function deAllocateFromNationalBank(uint256 amount, address nb) external;
function returnSurplusToParent() external;       // NEW: upward surplus
function setReserveRatio(uint256 ratioBps) external;
function getReserveRatio() external view returns (uint256);
function totalAllocated() external view returns (uint256);
function reserveBalance() external view returns (uint256);
function setOracleAddress(address oracle) external; // NEW: AI oracle wiring
function setTimeLock(address tl) external;          // NEW: TimeLock binding
function setKycVerifier(address kyc) external;      // NEW: KYC gate
function setAmlVerifier(address aml) external;      // NEW: AML gate
function pauseFunction(bytes32 fn) external;        // NEW: granular pause
function unpauseFunction(bytes32 fn) external;      // NEW: granular pause
function authorizeUpgrade(address newImpl) external; // UUPS gate
event ReserveAllocated(address indexed nb, uint256 amount);
event SurplusReturned(address indexed from, uint256 amount);
event ReserveRatioChanged(uint256 oldBps, uint256 newBps);
```

**Listing D.1:** WorldBankReserve public interface (extended with v14 functions).

**Table E.1:** Full MiCA / GENIUS Act compliance mapping.

| Regulatory rule | Description | CWB design choice | Status |
|---|---|---|---|
| MiCA Art. 30 – E-Money Token 1:1 backing | EMTs must be 100% backed by reserve assets | InsuranceFund + reserve-ratio enforcement; quarterly attestation | ○ |
| MiCA Art. 16 – Authorisation | EMT issuance requires CASP authorisation | Sandbox-first pilot via Bangladesh Bank RFFO; EU CASP through partner | ○ |
| MiCA Art. 30 – Reserve audit | Regular audit of reserve assets | Certora CVL + Tenderly + quarterly external audit | ○ |
| MiCA Art. 36 – Custody | Reserve custody segregated from operating funds | WorldBankReserve contract holds reserve; separate ops wallet | ○ |
| GENIUS Act §4 – Federal payment stablecoin | US framework for payment stablecoins | USDC integration via Circle licensed issuer | ○ |
| GENIUS Act §5 – Consumer protection | Mandatory redemption rights | SavingsVault withdraw guaranteed by reserve-ratio invariant | ○ |
| GENIUS Act §7 – Audit requirements | Monthly issuer attestation | Tenderly continuous monitor + monthly subgraph snapshot | ○ |
| Bangladesh FE Circular No. 06 | Electronic LC + blockchain banking communication | Letter-of-credit module (pre-thesis 2 + later phase) | ○ |
| Bangladesh Payment & Settlement Act 2024 | Payment service licensing | Local Bank operator partner licensed under the Act | ○ |
| GDPR / Bangladesh PDPA | Personal data protection | PII off-chain only; ZKP gate; encryption at rest in S3 | ○ |
| FATF Travel Rule | USD 1000+ tx counterparty info | ZKP-AML attestation + counterparty Local Bank metadata | ○ |

# Appendix E

# MiCA / GENIUS Act Compliance Mapping

# Appendix F

# Reproducibility Note

The thesis is reproducible from the public repository [Juboraz and Ahmed(2026)]:

1. Clone the repository and install macOS prerequisites: `brew install node@20 python@3.11 mactex`; `npm install -g @mermaid-js/mermaid-cli`.
2. Render improved diagrams: `bash Documentation/Diagrams/build-improved-diagrams.sh`.
3. Compile the thesis: `cd Documentation && pdflatex Pre-thesis_v14.tex && bibtex Pre-thesis_v14 && pdflatex Pre-thesis_v14.tex && pdflatex Pre-thesis_v14.tex`.
4. Run smart-contract tests: `cd contracts && npx hardhat test`.
5. Spin up the demo stack: `docker compose up`.

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